DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

Current version: 2026-04-17
Current number of commits: 905
Current git SHA1: 29e7c1c3c5c55c3b35f5c39b9461a36e10cf822f
Code lines: 12207 total lines, 11231 not empty lines, 976 empty lines.

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1 Files trees

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
```

```
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
```

```
│    │ an_see_that_it_works.txt
│    ├── funny.sh
│    ├── spampty_data.txt
│    ├── string_escaping.html
│    ├── string_escaping.xhtml
│    └──
│      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│      that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 83 files

[ 12K Apr 17 18:27]  ./
├── [4.0K Apr 17 18:26]  build_and_checks_dependencies/
│    ├── [4.4K Apr  1 19:39]  build_and_checks.exec.sh.tpl*
│    ├── [1.2K Mar 26 12:30]  build_dependencies_notes.exec.php*
│    ├── [8.3K Apr 17 09:32]  build_dependencies_notes.libr.php
│    ├── [4.2K Apr 17 10:34]  build_md_from_printable_md.exec.sh*
│    ├── [3.2K Apr 16 20:21]  check_shell_scripts_beginnings.libr.sh
│    ├── [5.3K Apr 17 10:11]  check_shell_scripts_indentation.libr.sh
│    ├── [3.7K Apr 17 10:15]  check_URLs.libr.sh
│    ├── [ 34K Apr 17 18:08]  common_build_and_checks.exec.sh*
│    ├── [ 29K Apr 17 18:08]  common_build_and_checks.exec.sh.tpl
│    ├── [ 18K Apr 17 11:22]  common_options.libr.sh
│    ├── [6.2K Apr 16 19:40]  comparisons.libr.sh
│    ├── [ 22K Apr 17 12:46]  create_PDF.exec.sh*
│    ├── [3.7K Apr 17 12:41]  generate_from_template.libr.sh
│    ├── [2.0K Mar 24 20:52]  get_common_text_glob_patterns.libr.sh
│    ├── [3.3K Apr 16 20:21]  grammar_and_spelling_check.libr.sh
│    ├── [4.0K Apr 17 12:17]  licenses_templates/
│    │    ├── [ 14K Apr 17 12:17]  build_licenses_templates.exec.sh*
│    │    ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│    │    ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│    │    └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│    ├── [6.0K Apr 16 20:21]  lines_counts.libr.sh
│    ├── [ 15K Apr 17 12:28]  lines_filters.libr.sh
│    ├── [3.0K Apr  1 14:56]  lines_maps.libr.sh
│    ├── [3.3K Apr 17 12:56]  overwrite_if_not_equal.libr.sh
│    ├── [2.4K Apr 16 20:21]  python_black_complement.libr.sh
│    ├── [3.1K Apr 17 13:13]  python_isort_complement.libr.sh
```

```
│   ├── [2.1K Apr 16 20:21]  repository_data.libr.sh
│   ├── [4.0K Apr 17 13:35]  shell_checks_complement.libr.sh
│   ├── [2.0K Mar 26 12:37]  split_score.exec.php
│   ├── [3.2K Mar 26 23:11]  split_score.libr.php
│   ├── [ 47K Apr 17 14:09]  strings_functions.libr.sh
│   ├── [4.6K Apr 17 11:42]  too_long_code_lines.libr.sh
│   ├── [4.5K Apr 17 14:13]  update_common_build_and_checks.exec.sh*
│   └── [6.0K Apr 17 14:18]  update_or_check_files_names_listing.exec.sh*
├── [4.6K Apr 17 18:08]  build_and_checks.exec.sh*
├── [4.0K Apr 17 18:27]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [5.1K Apr 17 16:50]  dependencies_data.json
│   ├── [4.3K Apr 17 16:57]  dependencies_notes.txt
│   ├── [5.7K Apr 17 16:51]  DevOrSysAdminScripts.html.tpl
│   ├── [ 35K Apr 17 18:09]  DevOrSysAdminScripts.tex
│   ├── [4.2K Apr 17 16:52]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.5K Mar 26 22:08]  files_names_listing.txt
│   ├── [  23 Mar 24 19:33]  .gitignore
│   ├── [1.4K Apr 17 16:49]  licenses_notes.txt
│   ├── [ 13K Apr 17 09:38]  phpcs_ruleset.xml
│   ├── [2.2K Apr 17 14:32]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.6K Apr 17 17:09]  repository_data.txt
│   ├── [ 841 Apr 17 17:08]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 166 Mar 24 19:33]  shellcheck.ini
│   └── [3.6K Apr  2 09:16]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [484K Apr 17 18:09]  DevOrSysAdminScripts.html
├── [416K Apr 17 18:09]  DevOrSysAdminScripts.pdf
├── [4.0K Apr 17 16:31]  git_helpers/
│   ├── [3.8K Apr 17 16:20]  git_checks.libr.sh
│   ├── [2.1K Apr  1 19:21]  post-commit*
│   ├── [ 15K Apr 17 16:31]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Apr  1 19:23]  pre-commit*
├── [   8 Mar 26 10:37]  .gitignore
├── [8.3K Apr 17 16:33]  IMAP_transfert.py
├── [1.1K Mar 31 19:59]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
```

```
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 01234\
    │── [   0 Mar  4 10:01]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345\
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456\
    │── [   0 Mar  4 09:13]
    │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
    │ an_see_that_it_works.txt
    │── [1.3K Mar  9 15:15]  funny.sh
    │── [  94 Mar  3 19:56]  spampty_data.txt
    │── [ 442 Mar 18 16:50]  string_escaping.html
    │── [ 654 Mar 18 17:27]  string_escaping.xhtml
    └── [   0 Mar  4 09:13]
      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
      that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [7.0K Apr 17 18:12]  string_escaping.libr.php
├── [ 17K Apr 17 18:22]  unicode.libr.php
├── [5.5K Apr 17 13:19]  upgrade_build_and_checks.exec.sh*
└── [2.5K Apr 16 19:36]  wget_sha512.libr.sh

6 directories, 83 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:

basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,

https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/,
https://docs.phpdoc.org/,
https://github.com/phpdocumentor/phpdocumentor.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home,
https://gitlab.gnome.org/GNOME/libxml2.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,

https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
```

```
  LFBFL_work_directory="$1"
fi
readonly LFBFL_work_directory

local LFBFL_verbose=""
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
fi
readonly LFBFL_verbose

local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download=y* ]]; then
  LFBFL_check_download="--check-download=y"
fi
readonly LFBFL_check_download

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
mkdir --parents "${LFBFL_subdir}/licenses_templates"
local LFBFL_variables_directory
LFBFL_variables_directory="${LFBFL_work_directory}/"
LFBFL_variables_directory+="build_and_checks_variables"
readonly LFBFL_variables_directory
mkdir --parents -- "${LFBFL_variables_directory}/temp"
if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
  printf "temp/\nupgrade_venvs_ts\n"\
    > "${LFBFL_variables_directory}/.gitignore"
fi

local LFBFL_dependencies_URL
LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
readonly LFBFL_dependencies_URL

declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
local LFBFL_script_download_URL
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
LFBFL_script_download_URL+="${LFBFL_common_file_name}"
readonly LFBFL_script_download_URL
declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
@sha512_common_build_and_checks.exec.sh@
wget_sha512 "${LFBFL_file_path}"\
```

```
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts library.

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@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

build_dependencies_notes($argv[1]);
?>
```

3.3 build_and_checks_dependencies/build_dependencies_notes.libr.php

```php
<?php

/**
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2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';
```

```
/**
This function looks at the dependencies data found in a work directory,
and generate three files: a txt file, a TeX file (.sub),
and an HTML file (.sub) with the data in a nicer format.

@param string $s_work_directory The directory where the work must be done.

@return void
*/
function build_dependencies_notes(string $s_work_directory) : void {
  $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  $arr_data = json_decode($s_data, true);
  $s_data_file_path2 = $s_subdir1.'/repository_data.txt';
  $s_data2 = file_get_contents($s_data_file_path2);
  $i_max_line_length = 70;
  foreach(explode("\n", $s_data2) as $s_line){
    if(str_starts_with($s_line, 'max_line_length=')){
      $i_max_line_length = (
        (int) substr($s_line, strlen('max_line_length='))
      );
      break;
    }
  }

  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    $s_URL_word = 'URL';
    if(count($arr_dependency['URLs']) > 1){
      $s_URL_word .= 's';
    }

    $s_result_content3 .= "<p>\n";

    $s_dependency1 = $arr_dependency['dependency_name'];
    $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
    $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);

    $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
    $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
    $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
```

```
    $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";

    for(
      $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
      $i <= $i_max;
      ++$i
    ){
      $mixed_URL = $arr_dependency['URLs'][$i];
      $s_comment = '';
      $s_URL = '';
      if(is_array($mixed_URL)){
        $s_URL = $mixed_URL['URL'];
        $s_comment = $mixed_URL['comment'];
      }
      else{
        $s_URL = $mixed_URL;
      }
      $i_URL_length = strlen($s_URL);
      $i_comment_length = strlen($s_comment);

      $s_result_content1 .= $s_URL;
      $i_previous_line_length = $i_URL_length;
      if($i_comment_length > 0){
        if(
          $i_previous_line_length + $i_comment_length + 1
          > $i_max_line_length
        ){
          $s_result_content1 .= "\n";
        }
        else{
          $s_result_content1 .= ' ';
        }
        $s_result_content1 .= $s_comment;
      }
      $s_result_content1 .= "\n";

      if($i_URL_length > 60){
        $s_result_content2 .= (
          "{\\small\n"
          .'  \url{'.$s_URL."}\n"
          .'}'
        );
        $i_previous_line_length = 1;
      }
      else{
        $s_result_content2 .= '\url{'.$s_URL.'}';
```

```
        $i_previous_line_length = $i_URL_length + 6;
      }
      if($i === $i_max){
        ++$i_previous_line_length;
      }
      else{
        $i_previous_line_length += 9;
      }
      if($i_comment_length > 0){
        if(
          $i_previous_line_length + $i_comment_length + 1
          > $i_max_line_length
        ){
          $s_result_content2 .= "\n";
        }
        else{
          $s_result_content2 .= ' ';
        }
        $s_result_content2 .= string_escaping\TeX\escape_text($s_comment);
      }
      if($i === $i_max){
        $s_result_content2 .= ".\n";
      }
      else{
        $s_result_content2 .= ",\\newline\n";
      }

      $s_result_content3 .= (
        "<a\n"
        .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
        ."><span style=\"font-family:monospace\"\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      if($s_comment !== ''){
        $s_result_content3 .= (
          ' '.string_escaping\HTML\escape_text($s_comment)
        );
      }
      if($i === $i_max){
        $s_result_content3 .= ".\n";
      }
      else{
        $s_result_content3 .= ",<br>\n";
      }
    }//end for urls
```

```
    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }//end foreach($arr_data as $arr_dependency)

  // Add a "footer".
  $s_closing_sentence = (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content1 .= $s_closing_sentence;
  $s_result_content2 .= $s_closing_sentence;
  $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";

  // We overwrite results if needed.
  $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
  $s_existing_result_content1 = '';
  if(file_exists($s_result_path1)){
    $s_existing_result_content1 = file_get_contents($s_result_path1);
  }
  if($s_result_content1 !== $s_existing_result_content1){
    file_put_contents($s_result_path1, $s_result_content1);
  }
  /*
  Example of generated content of dependencies_notes.txt:
  ---------------------------------------------------------------------------
  more of util-linux can be found at the following URLs:
  https://www.kernel.org/pub/linux/utils/util-linux/
  https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
  https://github.com/util-linux/util-linux official?
  ---------------------------------------------------------------------------
  */
  $s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
  $s_existing_result_content2 = '';
  if(file_exists($s_result_path2)){
    $s_existing_result_content2 = file_get_contents($s_result_path2);
  }
  if($s_result_content2 !== $s_existing_result_content2){
    file_put_contents($s_result_path2, $s_result_content2);
  }
  /*
  Example of generated content of dependencies_notes.tex.sub:
  ---------------------------------------------------------------------------
  more of util-linux can be found at the following URLs:\newline
  \url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
```

```
  {\small
    \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
  },\newline
  \url{https://github.com/util-linux/util-linux} official?.
  ---------------------------------------------------------------------------
  */
  $s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
  $s_existing_result_content3 = '';
  if(file_exists($s_result_path3)){
    $s_existing_result_content3 = file_get_contents($s_result_path3);
  }
  if($s_result_content3 !== $s_existing_result_content3){
    file_put_contents($s_result_path3, $s_result_content3);
  }
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

3.4 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name README README 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
    printf "Searching md file: %s.\n" "${LFBFL_base_name}"
  fi

```

```
# Remove line returns here to keep lines short.
local LFBFL_sed_expression
LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
readonly LFBFL_sed_expression
# Since Markdown can contain code, it is limited to Markdown URLs.
# Hence, it is way more complicated than 's/\\\n//Mg' expression
# in other files.
# Another downside is that more than one pass is needed.

if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
  sed --regexp-extended --null-data\
    --expression="${LFBFL_sed_expression}"\
    --expression="${LFBFL_sed_expression}"\
    --expression="${LFBFL_sed_expression}"\
    --expression="${LFBFL_sed_expression}"\
    -- "${LFBFL_base_name}.md.tpl"\
    > "${LFBFL_base_name}.md.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.md"\
    "${LFBFL_base_name}.md.temp"
else
  printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
  # Unfortunately, we cannot return early, since we want to propose
  # HTML file generation with pandoc from .md file,
  # if the file .md.tpl doesn't exist.
fi

if [[ ! -f "${LFBFL_base_name}.md" ]]; then
  printf "No file %s.md\n" "${LFBFL_base_name}"
  return 1
fi

pandoc --from=markdown --to=html --standalone\
  --shift-heading-level-by=-1 --output="${LFBFL_base_name}.html.temp"\
  -- "${LFBFL_base_name}.md"
# Correcting pandoc HTML "by hand".
# https://github.com/jgm/pandoc/issues/10957
# With the most open-minded answer found on Internet ;) XD.
# This second sed expression is what I need,
# but it may be unsuitable for you.
# Other problems:
# https://github.com/jgm/pandoc/issues/11484
local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
readonly LFBFL_prewrap
```

```
  sed --in-place --regexp-extended\
    --expression='s~(  <meta .*) />~\1>~g'\
    --expression='s~<html .*>~<html lang="en">~'\
    --expression='/    code\{white-space: pre-wrap;\}/d'\
    -- "${LFBFL_base_name}.html.temp"
  sed --in-place --regexp-extended --null-data\
    --expression='s~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
    --expression="${LFBFL_prewrap}"\
    -- "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.5 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

```

```
LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename -- "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head --lines=1 -- "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
```

```
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head --lines=1 -- "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

3.6 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.libr.sh
27 source "./${LFBFL_subdir}/common_options.libr.sh"
28 # shellcheck source=lines_filters.libr.sh
29 source "./${LFBFL_subdir}/lines_filters.libr.sh"
30
31 check_one_shell_script_indentation(){
32   # $1=file_path
33   declare -r LFBFL_NULL_PREVIOUS_LINE="--"
34   declare -i LFBFL_i_file_with_error=0
35   local LFBFL_some_line
36   local LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
37   local LFBFL_line_end
38   local LFBFL_previous_line_end
39   declare -i LFBFL_i_line_length
40   declare -i LFBFL_i_line_end_length
41   declare -i LFBFL_i_line_start_length
42   local LFBFL_spacing
43   declare -i LFBFL_i_offset
44   local LFBFL_substring1
45   local LFBFL_substring2
46   declare -r LFBFL_pattern='^ *-e '
47
48   # This grep would have been enough without some sed commands.
49   declare -r LFBFL_s_lines=$(
50     grep --before-context=1 --extended-regexp --line-number\
51       --regexp='^(  )* ([^ ]|$)' --with-filename -- "$1"
52   )
53   if [[ -z "${LFBFL_s_lines}" ]]; then
54     return
55   fi
56
57   declare -a LFBFL_arr_lines
58   mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
59   readonly LFBFL_arr_lines
60
61   for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
62     if [[ "${LFBFL_some_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]]; then
63       # When switching between code fragments, we reinit.
64       # It cannot happen for other grep output lines with line-number.
65       LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
66       continue
67     fi
68     # Note how it's funny that this loop always work because we check
69     # shell scripts beginnings in another script and thus we always
```

```
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number.
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

```

```
  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.7 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
50
51    declare -Ar LFBFL_substitutions=(
52      ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
53    )
54
55    local LFBFL_pattern
56    local LFBFL_base_file_name
57    local LFBFL_substitution
58    local LFBFL_substitution2
59    local LFBFL_file_path
60    local LFBFL_s_files_paths
61    declare -a LFBFL_arr_files_paths
62    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
63      if [[ LFBFL_i_verbose -eq 1 ]]; then
64        printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
65      fi
66      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
67        | xargs grep --with-filename -- 'http:'\
68        | grep --invert-match "| xargs grep --with-filename -- 'htt"\
69        | grep --invert-match --perl-regexp\
70          "(['\"])http(:[^'\"]*)\\1.*(['\"])https\\2\\3"\
71        | grep --invert-match 'http://www.w3.org/1999/xhtml'\
72        | grep --invert-match\
73          'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
74      # Second grep after xargs grep just above will remove false positives
75      # from substitutions that fit on one line.
76
77      LFBFL_s_files_paths=$(
78        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
79        | relevant_find\
80        | not_main_html_find
81      )
82      if [[ -z "${LFBFL_s_files_paths}" ]]; then
83        [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
84        continue
85      fi
86      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
87      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
88        LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
89        [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
90          || continue
91        if [[ LFBFL_i_verbose -eq 1 ]]; then
92          printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
93        fi
94        for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
95          LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
```

```
        if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
        then
          sed --in-place\
          --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            -- "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
```

```
34  # $1 LFBFL_work_directory
35  # $2 LFBFL_dependencies_URL
36  # Options:
37  #   --check-download=y|keep (=keep to keep the downloaded files)
38  #   --verbose
39  local LFBFL_work_directory="${1:-.}"
40  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41  declare -r LFBFL_dependencies_URL="$2"
42  local LFBFL_verbose=""
43  declare -i LFBFL_i_verbose=0
44  if [[ "$*" == *--verbose* ]]; then
45    printf "%s %s\n" "$0" "$*"
46    LFBFL_verbose="--verbose"
47    LFBFL_i_verbose=1
48  fi
49  readonly LFBFL_verbose
50  readonly LFBFL_i_verbose
51  local LFBFL_check_download=""
52  if [[ "$*" == *--check-download=keep* ]]; then
53    LFBFL_check_download="--check-download=keep"
54  elif [[ "$*" == *--check-download=y* ]]; then
55    LFBFL_check_download="--check-download=y"
56  fi
57  readonly LFBFL_check_download
58
59  declare -a LFBFL_some_common_options=(
60    "${LFBFL_verbose}"
61    "--work-directory=${LFBFL_work_directory}"
62  )
63
64  source ./wget_sha512.libr.sh
65
66  local LFBFL_subdir="build_and_checks_dependencies"
67  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68  declare -r LFBFL_subdir3="build_and_checks_variables"
69  local LFBFL_file_name
70  local LFBFL_script_download_URL
71  local LFBFL_file_path
72  local LFBFL_correct_sha512
73
74  wrapped_wget_sha512(){
75    wget_sha512 "${LFBFL_file_path}"\
76      "${LFBFL_script_download_URL}"\
77      "${LFBFL_correct_sha512}"\
78      "${LFBFL_verbose}"\
79      "${LFBFL_check_download}"
```

```
80      }
81
82      LFBFL_file_name="build_dependencies_notes.exec.php"
83      LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85      LFBFL_correct_sha512='624368d161351d7a261fbe3165d51907f46394c1b0e79'
86      LFBFL_correct_sha512+='b980ec11173c324b2c5531cf2a2b60b2dc62edcaa9c8'
87      LFBFL_correct_sha512+='71dca55065f8802df4275420836252786f9606b'
88      wrapped_wget_sha512
89      chmod +x "./${LFBFL_file_path}"
90
91      LFBFL_file_name="build_dependencies_notes.libr.php"
92      LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
93      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
94      LFBFL_correct_sha512='49e9469e44e3a18db98810ab734b89ff1268e7a9c7788'
95      LFBFL_correct_sha512+='12e8fb853aff5a26aaf00845975c73fd7cf6ef5011aa'
96      LFBFL_correct_sha512+='462629ca4c6d8db119668f429c956c040b07894'
97      wrapped_wget_sha512
98
99      LFBFL_file_name="build_md_from_printable_md.exec.sh"
100     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
101     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
102     LFBFL_correct_sha512='4ed727f21bfdb99d3c3a37d631d27816da10be357a200'
103     LFBFL_correct_sha512+='b03516cc642cc720735bd1363e47e1b38d5a4667a4ef'
104     LFBFL_correct_sha512+='75c573e92f09261df6c2c55ee9c0852ab2d1ee8'
105     wrapped_wget_sha512
106     chmod +x "./${LFBFL_file_path}"
107
108     LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
109     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
110     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
111     LFBFL_correct_sha512='beb8e11e2ac7fa636a0ac586f2378d8c8f72b16320a72'
112     LFBFL_correct_sha512+='99533bd4fbcdbd8794281d78a8882dee54eb53d639a9'
113     LFBFL_correct_sha512+='0accdf8f557ac6f64bd9b9d6fe3eae9d4789f54'
114     wrapped_wget_sha512
115
116     LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
117     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
118     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
119     LFBFL_correct_sha512='9ebc5cfab69e6beab5e77495b2585ce9bc3ad84af4e9c'
120     LFBFL_correct_sha512+='20f5a8ed390730fbeb0bf13ff4cfef635dbdbfff8872'
121     LFBFL_correct_sha512+='8b1fe95595ef4d47a7f535cd4644c0ccbfa384c'
122     wrapped_wget_sha512
123
124     LFBFL_file_name="check_URLs.libr.sh"
125     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='9e42d111550432b63d63f9faa35c92c45a9e4ef308a65'
LFBFL_correct_sha512+='1ecf094a0838ac36fbe36109b3dc9b4d48a24887d155'
LFBFL_correct_sha512+='dae7b546ea64551493d0dc4e772e1b09248fa4e'
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='10505b7876c808bc24478ba2a4c1ba530375abc0121f6'
LFBFL_correct_sha512+='ed742a1de5e220369810360ffb1b5ab03892e2277438'
LFBFL_correct_sha512+='78caff0e57696805b0099aad8022720413b46c6'
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='27a77b03184fa1497ad0448206a85cfc85071d96364b9'
LFBFL_correct_sha512+='c7ea6bedcd956eecf14115f5f001d6d859e22caa690b'
LFBFL_correct_sha512+='3eebe5805afaa94e530ef8de01d54da408956c3'
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='560add021e3717e42fa27c8ad408e53974382259ecdca'
LFBFL_correct_sha512+='73e3f128fbe65cdb9fd3aa8b0dd1dbd329c39b9232ee'
LFBFL_correct_sha512+='9eafa1bacf1b9b4cb00cc305c79ae857a62fdd3'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='f983105225e806681b004aeb73d5bdd4d2bd8bd0d71ab'
LFBFL_correct_sha512+='9b6294c76252302b59f119b944185ffec94fa87b2429'
LFBFL_correct_sha512+='6916a5be393ba987d7464c2871bb448039f85d0'
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='d38b894f63ac0db1eab4447a765b723642e12e3443aee'
LFBFL_correct_sha512+='01bd08e44bd2971bf11d3f0b98d2932bcaaeab4f30b7'
LFBFL_correct_sha512+='9f7da666e02a0fb1dada8a348ad866a104e4d53'
wrapped_wget_sha512
```

```
172
173     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
174     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
175     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
176     LFBFL_correct_sha512='5722273402a4e8afbae4bc10e7a50f3d74d942404df18'
177     LFBFL_correct_sha512+='bc59f1bb12cea917f6292ee3dd13a7533d16fd03af99'
178     LFBFL_correct_sha512+='c4941df14027cce33afae1c0db169f15e503f2c'
179     wrapped_wget_sha512
180
181     # /licenses_templates/ ---------------------------------------------
182     local LFBFL_dependencies_URL2
183     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
184     readonly LFBFL_dependencies_URL2
185     LFBFL_file_name="build_licenses_templates.exec.sh"
186     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
187     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
188     LFBFL_correct_sha512='72975aaa23962f7c110a418e6d3df28854ea9037919ea'
189     LFBFL_correct_sha512+='1f0db8a538ef8836c8324368a03351a1ba357bd6b9a7'
190     LFBFL_correct_sha512+='4e7ea75adfbac0df616a36663e57ef98d325e72'
191     wrapped_wget_sha512
192     chmod +x "./${LFBFL_file_path}"
193
194     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
195     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
196     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
197     LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
198     LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
199     LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
200     wrapped_wget_sha512
201
202     LFBFL_file_name="license_file_header_GPLv3+.tpl"
203     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
204     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
205     LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
206     LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
207     LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
208     wrapped_wget_sha512
209
210     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
211     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
212     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
213     LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
214     LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
215     LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
216     wrapped_wget_sha512
217     # ------------------------------------------------------------------
```

```
218
219     LFBFL_file_name="lines_counts.libr.sh"
220     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
221     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
222     LFBFL_correct_sha512='97e50d1fd31841c8e5b747503a55999d5bd3d394808c3'
223     LFBFL_correct_sha512+='037b321ab54057518a861e4bb97fa7659aee061a31cf'
224     LFBFL_correct_sha512+='e30ac8af22f268f19aa51accf334cbc38aa9091'
225     wrapped_wget_sha512
226
227     LFBFL_file_name="lines_filters.libr.sh"
228     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
229     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
230     LFBFL_correct_sha512='76050708d970c346dadfb46599ec32c26fc831990b163'
231     LFBFL_correct_sha512+='3437360260ca11ae6b2c4784f5ff1331cfbc70e19ef0'
232     LFBFL_correct_sha512+='0e38184448aa57f053270254582cc1319ea0baa'
233     wrapped_wget_sha512
234
235     LFBFL_file_name="lines_maps.libr.sh"
236     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
237     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
238     LFBFL_correct_sha512='45df460ebb31fd5f2f2a62a70f03d6ab9fd3bfd8d6e6f'
239     LFBFL_correct_sha512+='e45650e339f5b6bad5557a8c8d3a830d8539ba5442df'
240     LFBFL_correct_sha512+='68945da69270219159ceea45b14b395793fc025'
241     wrapped_wget_sha512
242
243     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
244     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
245     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
246     LFBFL_correct_sha512='8c004f2256257e4c47e96660fd4a619d5fd1996d709ff'
247     LFBFL_correct_sha512+='dcecf810f0f607544fc8f080d2f528de28f992293958'
248     LFBFL_correct_sha512+='bc189d1db0fcab98f9199c2a2df39722a8e2473'
249     wrapped_wget_sha512
250
251     LFBFL_file_name="python_black_complement.libr.sh"
252     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
253     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
254     LFBFL_correct_sha512='1511766231f7463236f9eb31da045b7cf261f356de956'
255     LFBFL_correct_sha512+='01431a015d8fbac9fd823a1ac5c2ffd16baedc2228c8'
256     LFBFL_correct_sha512+='6c78bbec73d934ef2a119ae5e1416963cbdb1f9'
257     wrapped_wget_sha512
258
259     LFBFL_file_name="python_isort_complement.libr.sh"
260     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
261     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
262     LFBFL_correct_sha512='f98c63d14f74ebe1cf702e0c9107dd1a4add29d4c4f62'
263     LFBFL_correct_sha512+='37e8d6e11c9d0454fc2e4ad13a0cb66ececf00a02d2a'
```

```
LFBFL_correct_sha512+='39004860e648fe7f49a82f2a7b59d844e37bc25'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='b69cffe97c11ae5d98532f65e61c94139d781dab2c859'
LFBFL_correct_sha512+='985b4bf641a1f846337ecd7b02fbf823551934e80b46'
LFBFL_correct_sha512+='0c9dc4ca6fa9b4ca07facfe3b1a8b239ea29501'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='745109db3fbd3d20b60427a2e116b577eca1c0049e532'
LFBFL_correct_sha512+='7f53a925b0814dd9f98b5b43986a8b4367f80e1189ae'
LFBFL_correct_sha512+='ffca99ff15afdf37b062fb48cd73315a4881cf4'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='0eaca0166e71a1992f897c7f87e77da3babd297cd85c6'
LFBFL_correct_sha512+='e4251ae920f15b4cbd9c3d4a7e2ae21eaf74b366a0e7'
LFBFL_correct_sha512+='681977dff58901c6d40b3e46f38e0cbec86be54'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='47b667d83125fa17babd46201ae8a8b7f7abeef123276'
LFBFL_correct_sha512+='40e7ed87faeacac997455207c4ab4df2734649676960'
LFBFL_correct_sha512+='1efcadaf35338ba2bec2410e01065ff7ed871f7'
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='a00516fb9ed9b12127d6c2d8626b47027ef7704123475'
LFBFL_correct_sha512+='2f37ef54babc086078250c2b330111754913d6671913'
LFBFL_correct_sha512+='9a3f065be1363538da331c756072a98036e600d'
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='ca49414202ff1bbd7a607d4404e7f92494fe45e150d95'
LFBFL_correct_sha512+='d9ad0aded752c8ca8e3f8c4509823cd462fb481a09f5'
LFBFL_correct_sha512+='fe6c90be9d8af7e4411afa84429bb87b7ea6c41'
wrapped_wget_sha512

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='609d8f15c69f9d766c1435e0dfb3571b4329ab09a29ae'
LFBFL_correct_sha512+='087c4753f7692809873e94220d0c71506b1592134431'
LFBFL_correct_sha512+='124ab42ecf703ed60486d1273d35a5cc2172c71'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
```

```
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
    LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
```

```
402         --work-directory="${LFBFL_some_directory}"\
403         --base-name="${LFBFL_file_name}"\
404         "${LFBFL_verbose}"
405       done
406     fi
407
408     printf "Building dependencies_notes\n"
409     # https://github.com/php/php-src/issues/21538
410     local LFBFL_annoying_warning
411     LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
412     LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
413     LFBFL_annoying_warning=" by settings in"
414     readonly LFBFL_annoying_warning
415     php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
416       -- "${LFBFL_work_directory}"\
417       2>&1\
418       | grep --invert-match "${LFBFL_annoying_warning}"
419
420     declare -i LFBFL_i_directory_changed
421     pushd_to_work_directory
422     LFBFL_i_directory_changed=$?
423     can_continue_after_enhanced_pushd || return 1
424
425     local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
426
427     declare -i LFBFL_i_max_line_length
428     grep_variable "${LFBFL_data_file_path}" max_line_length\
429       --result-variable-prefix=LFBFL_i_
430
431     LFBFL_some_common_options+=(
432       "--max-line-length=${LFBFL_i_max_line_length}"
433     )
434     LFBFL_some_common_options2+=(
435       "--max-line-length=${LFBFL_i_max_line_length}"
436     )
437
438     declare -i LFBFL_i_upgrade_venvs=0
439     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
440     readonly LFBFL_upgrade_venvs_ts_file
441     declare -i LFBFL_i_upgrade_venvs_ts
442     declare -i LFBFL_i_current_ts
443     local LFBFL_upgrade_venvs_answer
444
445     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
446     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
447       "${LFBFL_some_common_options2[@]}"
```

```

if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
  LFBFL_i_upgrade_venvs_ts=$(
    stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
  )
  LFBFL_i_current_ts=$(date +%s)
  ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
  ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
  if [[ LFBFL_i_current_ts -gt 0 ]]; then
    LFBFL_i_upgrade_venvs=1
  fi
else
  LFBFL_i_upgrade_venvs=1
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  local LFBFL_upgrade_venvs=""
  grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
    --result-variable-prefix=LFBFL_
  if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
    read -r -n 1 -t 10\
      -p "Upgrade venvs and composer global? [Y/n]"\
      LFBFL_upgrade_venvs_answer
    if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
      LFBFL_i_upgrade_venvs=0
    fi
  fi
fi

printf "Analyzing too long lines\n"
too_long_code_lines "${LFBFL_some_common_options2[@]}"

printf "Analyzing URLs\n"
check_URLs "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing strange characters: hover over in doubt\n"
# shellcheck disable=SC1111
LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘’└├│«»..."
grep --binary-files=without-match --color=always --exclude-dir .git\
  --line-number --perl-regexp --recursive --\
  "[^${LFBFL_usual_characters}]" .\
  | relevant_grep

printf "Grammar and spelling check\n"
grammar_and_spelling_check "${LFBFL_data_file_path}"\
  "${LFBFL_some_common_options2[@]}"
```

```
494
495   printf "Running shellcheck\n"
496   LFBFL_s_files_paths=$(
497     find . -type f -name "*.sh"\
498     | relevant_find
499   )
500   if [[ -n "${LFBFL_s_files_paths}" ]]; then
501     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
502     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
503       shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
504         -- "${LFBFL_file_path}"
505     done
506   fi
507   printf "Running shell_checks_complement\n"
508   shell_checks_complement "${LFBFL_some_common_options2[@]}"
509
510   printf "Analyzing shell scripts beginnings\n"
511   check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
512     | relevant_grep
513
514   printf "Analyzing shell scripts indentation\n"
515   check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
516     | relevant_grep
517
518   printf -- "---Python---\n"
519
520   maybe_deactivate(){
521     if command -v deactivate
522     then
523       deactivate
524     fi
525   }
526
527   source_without_return_trap(){
528     declare -i LFBFL_i_no_return_traps=1
529     # shellcheck disable=SC1090,SC1091
530     source -- "$1"
531   }
532
533   printf "Running isort\n"
534   local LFBFL_isort_venv=""
535   grep_variable "${LFBFL_data_file_path}" isort_venv\
536     --result-variable-prefix=LFBFL_\
537     --replace-line-returns-by=""
538   if [[ -n "${LFBFL_isort_venv}" ]]; then
539     maybe_deactivate
```

```
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed --regexp-extended --expression="${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_path}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
```

```
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_path}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
```

```
632   # Saving new baseline in temp if necessary.
633   bandit --ini="${LFBFL_subdir3}/bandit.ini"\
634     --format=json\
635     --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
636     --recursive .
637   if [[ -n "${LFBFL_bandit_venv}" ]]; then
638     deactivate
639   fi
640
641   printf "Running pylint\n"
642   LFBFL_pylint_venv=""
643   grep_variable "${LFBFL_data_file_path}" pylint_venv\
644     --result-variable-prefix=LFBFL_\
645     --replace-line-returns-by=""
646   if [[ -n "${LFBFL_pylint_venv}" ]]; then
647     maybe_deactivate
648     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
649   fi
650   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
651     pip install --upgrade pylint
652   fi
653   pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
654   if [[ -n "${LFBFL_pylint_venv}" ]]; then
655     deactivate
656   fi
657
658   printf "Running ruff\n"
659   LFBFL_ruff_venv=""
660   grep_variable "${LFBFL_data_file_path}" ruff_venv\
661     --result-variable-prefix=LFBFL_\
662     --replace-line-returns-by=""
663   if [[ -n "${LFBFL_ruff_venv}" ]]; then
664     maybe_deactivate
665     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
666   fi
667   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
668     pip install --upgrade ruff
669   fi
670   ruff check --config="${LFBFL_subdir3}/ruff.toml"
671   if [[ -n "${LFBFL_ruff_venv}" ]]; then
672     deactivate
673   fi
674   printf -- "---Python end---\n"
675
676   printf -- "---PHP---\n"
677   printf "Running PHP native linter\n"
```

```
LFBFL_s_files_paths=$(
  find . -type f -name "*.php"\
  | relevant_find
)
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    php --syntax-check -f="${LFBFL_file_path}"
  done
fi

printf "Running PHP_CodeSniffer\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require squizlabs/php_codesniffer
fi
phpcs --report=code\
  --standard=build_and_checks_variables/phpcs_ruleset.xml

printf "Running PHPMD\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require phpmd/phpmd
fi

declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file="${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
sed --in-place --expression='s/" file="/"\n    file="/g'\
  "${LFBFL_temp_phpmd_baseline}"
sed --in-place --regexp-extended\
  --expression='s~(file=".*)/>~\1\n  />~g'\
  "${LFBFL_temp_phpmd_baseline}"
diff --suppress-common-lines\
  "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
```

```
    | grep --invert-match --regexp='<!-- ' --regexp='^[0-9]\+d[0-9]\+$'

  printf "Running phpDocumentor\n"
  # shellcheck disable=SC2312
  docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  local LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  printf "Running xmllint\n"
  declare -r LFBFL_s_files_for_xmllint=$(
    find . -type f -iregex ".*\.\(xml\|xhtml\)"\
    | relevant_find\
    | sort
  )
  if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
    declare -a LFBFL_arr_files_for_xmllint
```

```
    mapfile -t LFBFL_arr_files_for_xmllint\
      <<< "${LFBFL_s_files_for_xmllint}"
    readonly LFBFL_arr_files_for_xmllint
    local LFBFL_xmllint_files_without_DTD=""
    grep_variable "${LFBFL_data_file_path}" xmllint_files_without_DTD\
      --result-variable-prefix=LFBFL_
    declare -a LFBFL_arr_xmllint_files_without_DTD
    if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
      mapfile -t LFBFL_arr_xmllint_files_without_DTD\
        <<< "${LFBFL_xmllint_files_without_DTD}"
    fi
    for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "xmllint on %s\n" "${LFBFL_file_path}"
      # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
      # xmllint --pedantic --noout -- "${LFBFL_file_path}"
      xmllint --pedantic --noout "${LFBFL_file_path}"
    done

    local LFBFL_file_path2
    for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
      for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
      do
        if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
          LFBFL_file_path=""
        fi
      done
      if [[ "${LFBFL_file_path}" == "" ]]; then
        continue
      fi
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "xmllint on %s\n" "${LFBFL_file_path}"
      xmllint --pedantic --noout --valid "${LFBFL_file_path}"
    done
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

```

```
816     if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
817     then
818       LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
819     fi
820
821     declare -i LFBFL_i_vnu_jar_ts
822     local LFBFL_vnu_jar_path=""
823     grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
824       --result-variable-prefix=LFBFL_\
825       --replace-line-returns-by=""
826     if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
827       LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
828       LFBFL_i_current_ts=$(date +%s)
829       ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
830       ((
831         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
832       ))
833       if [[ LFBFL_i_current_ts -gt 0 ]]; then
834         printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
835         printf "Go get the latest version here:\n"
836         printf "https://github.com/validator/validator/releases\n"
837       fi
838       # vnu has false positives for xhtml.
839       local LFBFL_use_vnu_for_xhtml=""
840       grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
841         --result-variable-prefix=LFBFL_
842       local LFBFL_find_regexp
843       if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
844         LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
845       else
846         LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
847       fi
848       declare -r LFBFL_s_files_for_Nu=$(
849         find . -type f -iregex "${LFBFL_find_regexp}"\
850         | relevant_find
851       )
852       if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
853         declare -a LFBFL_arr_files_for_Nu
854         mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
855         readonly LFBFL_arr_files_for_Nu
856         java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
857           --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
858       fi
859     fi
860     # ------------------------------------------------------------------
861
```

```
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.9 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
```

```
38  #   --verbose
39  local LFBFL_work_directory="${1:-.}"
40  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41  declare -r LFBFL_dependencies_URL="$2"
42  local LFBFL_verbose=""
43  declare -i LFBFL_i_verbose=0
44  if [[ "$*" == *--verbose* ]]; then
45    printf "%s %s\n" "$0" "$*"
46    LFBFL_verbose="--verbose"
47    LFBFL_i_verbose=1
48  fi
49  readonly LFBFL_verbose
50  readonly LFBFL_i_verbose
51  local LFBFL_check_download=""
52  if [[ "$*" == *--check-download=keep* ]]; then
53    LFBFL_check_download="--check-download=keep"
54  elif [[ "$*" == *--check-download=y* ]]; then
55    LFBFL_check_download="--check-download=y"
56  fi
57  readonly LFBFL_check_download
58
59  declare -a LFBFL_some_common_options=(
60    "${LFBFL_verbose}"
61    "--work-directory=${LFBFL_work_directory}"
62  )
63
64  source ./wget_sha512.libr.sh
65
66  local LFBFL_subdir="build_and_checks_dependencies"
67  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68  declare -r LFBFL_subdir3="build_and_checks_variables"
69  local LFBFL_file_name
70  local LFBFL_script_download_URL
71  local LFBFL_file_path
72  local LFBFL_correct_sha512
73
74  wrapped_wget_sha512(){
75    wget_sha512 "${LFBFL_file_path}"\
76      "${LFBFL_script_download_URL}"\
77      "${LFBFL_correct_sha512}"\
78      "${LFBFL_verbose}"\
79      "${LFBFL_check_download}"
80  }
81
82  LFBFL_file_name="build_dependencies_notes.exec.php"
83  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.exec.php@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.libr.php@
wrapped_wget_sha512

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
```

```
130     wrapped_wget_sha512
131
132     LFBFL_file_name="create_PDF.exec.sh"
133     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
134     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
135     @sha512_create_PDF.exec.sh@
136     wrapped_wget_sha512
137     chmod +x "./${LFBFL_file_path}"
138
139     LFBFL_file_name="generate_from_template.libr.sh"
140     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
141     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
142     @sha512_generate_from_template.libr.sh@
143     wrapped_wget_sha512
144
145     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
146     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
147     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
148     @sha512_get_common_text_glob_patterns.libr.sh@
149     wrapped_wget_sha512
150
151     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
152     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
153     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
154     @sha512_grammar_and_spelling_check.libr.sh@
155     wrapped_wget_sha512
156
157     # /licenses_templates/ ---------------------------------------------
158     local LFBFL_dependencies_URL2
159     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
160     readonly LFBFL_dependencies_URL2
161     LFBFL_file_name="build_licenses_templates.exec.sh"
162     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
163     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
164     @sha512_build_licenses_templates.exec.sh@
165     wrapped_wget_sha512
166     chmod +x "./${LFBFL_file_path}"
167
168     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
169     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
170     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
171     @sha512_license_file_header_AGPLv3+.tpl@
172     wrapped_wget_sha512
173
174     LFBFL_file_name="license_file_header_GPLv3+.tpl"
175     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wrapped_wget_sha512
```

```
222
223     LFBFL_file_name="repository_data.libr.sh"
224     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
225     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
226     @sha512_repository_data.libr.sh@
227     wrapped_wget_sha512
228
229     LFBFL_file_name="shell_checks_complement.libr.sh"
230     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
231     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
232     @sha512_shell_checks_complement.libr.sh@
233     wrapped_wget_sha512
234
235     LFBFL_file_name="split_score.exec.php"
236     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
237     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
238     @sha512_split_score.exec.php@
239     wrapped_wget_sha512
240
241     LFBFL_file_name="split_score.libr.php"
242     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
243     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
244     @sha512_split_score.libr.php@
245     wrapped_wget_sha512
246
247     LFBFL_file_name="strings_functions.libr.sh"
248     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
249     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
250     @sha512_strings_functions.libr.sh@
251     wrapped_wget_sha512
252
253     LFBFL_file_name="too_long_code_lines.libr.sh"
254     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
255     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
256     @sha512_too_long_code_lines.libr.sh@
257     wrapped_wget_sha512
258
259     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
260     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
261     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
262     @sha512_update_or_check_files_names_listing.exec.sh@
263     wrapped_wget_sha512
264     chmod +x "./${LFBFL_file_path}"
265
266     # shellcheck source=check_shell_scripts_beginnings.libr.sh
267     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
```

```
268     # shellcheck source=check_shell_scripts_indentation.libr.sh
269     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
270     # shellcheck source=check_URLs.libr.sh
271     source "./${LFBFL_subdir}/check_URLs.libr.sh"
272     # shellcheck source=common_options.libr.sh
273     source "./${LFBFL_subdir}/common_options.libr.sh"
274     # shellcheck source=comparisons.libr.sh
275     source "./${LFBFL_subdir}/comparisons.libr.sh"
276     # shellcheck source=generate_from_template.libr.sh
277     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
278     # shellcheck source=get_common_text_glob_patterns.libr.sh
279     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
280     # shellcheck source=grammar_and_spelling_check.libr.sh
281     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
282     # shellcheck source=lines_counts.libr.sh
283     source "./${LFBFL_subdir}/lines_counts.libr.sh"
284     # shellcheck source=lines_filters.libr.sh
285     source "./${LFBFL_subdir}/lines_filters.libr.sh"
286     # shellcheck source=lines_maps.libr.sh
287     source "./${LFBFL_subdir}/lines_maps.libr.sh"
288     # shellcheck source=overwrite_if_not_equal.libr.sh
289     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
290     # shellcheck source=python_black_complement.libr.sh
291     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
292     # shellcheck source=python_isort_complement.libr.sh
293     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
294     # shellcheck source=shell_checks_complement.libr.sh
295     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
296     # shellcheck source=strings_functions.libr.sh
297     source "./${LFBFL_subdir}/strings_functions.libr.sh"
298     # shellcheck source=too_long_code_lines.libr.sh
299     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
300
301     enhanced_set_bash_option extglob
302     # shellcheck source=repository_data.libr.sh
303     source "./${LFBFL_subdir}/repository_data.libr.sh"
304     # shellcheck disable=SC2154,SC2309
305     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
306       && enhanced_unset_bash_option extglob
307
308     declare -a LFBFL_return_traps_stack
309     init_return_trap
310
311     enhanced_set_shell_option pipefail --trap-unset
312
313     declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
```

```
  keep_options LFBFL_some_common_options2 --verbose

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    --work-directory="${LFBFL_work_directory}"\
    --base-name=README\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
      LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  # https://github.com/php/php-src/issues/21538
  local LFBFL_annoying_warning
  LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
  LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
  LFBFL_annoying_warning=" by settings in"
  readonly LFBFL_annoying_warning
  php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    -- "${LFBFL_work_directory}"\
    2>&1\
```

```
360       | grep --invert-match "${LFBFL_annoying_warning}"
361
362     declare -i LFBFL_i_directory_changed
363     pushd_to_work_directory
364     LFBFL_i_directory_changed=$?
365     can_continue_after_enhanced_pushd || return 1
366
367     local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
368
369     declare -i LFBFL_i_max_line_length
370     grep_variable "${LFBFL_data_file_path}" max_line_length\
371       --result-variable-prefix=LFBFL_i_
372
373     LFBFL_some_common_options+=(
374       "--max-line-length=${LFBFL_i_max_line_length}"
375     )
376     LFBFL_some_common_options2+=(
377       "--max-line-length=${LFBFL_i_max_line_length}"
378     )
379
380     declare -i LFBFL_i_upgrade_venvs=0
381     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
382     readonly LFBFL_upgrade_venvs_ts_file
383     declare -i LFBFL_i_upgrade_venvs_ts
384     declare -i LFBFL_i_current_ts
385     local LFBFL_upgrade_venvs_answer
386
387     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
388     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
389       "${LFBFL_some_common_options2[@]}"
390
391     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
392       LFBFL_i_upgrade_venvs_ts=$(
393         stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
394       )
395       LFBFL_i_current_ts=$(date +%s)
396       ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
397       ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
398       if [[ LFBFL_i_current_ts -gt 0 ]]; then
399         LFBFL_i_upgrade_venvs=1
400       fi
401     else
402       LFBFL_i_upgrade_venvs=1
403     fi
404     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
405       local LFBFL_upgrade_venvs=""
```

```
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Analyzing too long lines\n"
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘’└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        -- "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

```

```
452   printf "Analyzing shell scripts beginnings\n"
453   check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
454     | relevant_grep
455
456   printf "Analyzing shell scripts indentation\n"
457   check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
458     | relevant_grep
459
460   printf -- "---Python---\n"
461
462   maybe_deactivate(){
463     if command -v deactivate
464     then
465       deactivate
466     fi
467   }
468
469   source_without_return_trap(){
470     declare -i LFBFL_i_no_return_traps=1
471     # shellcheck disable=SC1090,SC1091
472     source -- "$1"
473   }
474
475   printf "Running isort\n"
476   local LFBFL_isort_venv=""
477   grep_variable "${LFBFL_data_file_path}" isort_venv\
478     --result-variable-prefix=LFBFL_\
479     --replace-line-returns-by=""
480   if [[ -n "${LFBFL_isort_venv}" ]]; then
481     maybe_deactivate
482     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
483   fi
484   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
485     pip install --upgrade isort
486   fi
487   isort --profile=black .
488   if [[ -n "${LFBFL_isort_venv}" ]]; then
489     deactivate
490   fi
491   printf "Running python_isort_complement\n"
492   python_isort_complement "${LFBFL_some_common_options2[@]}"
493
494   printf "Running black\n"
495   # First, we update the configuration file with max_line_length.
496   local LFBFL_update_max_length="s/^line-length = [0-9]*$"
497   LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
```

```
  sed --regexp-extended --expression="${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_path}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
```

```
544         printf "Running mypy\n"
545         LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
546         mypy -- "${LFBFL_directory_path}"
547       fi
548       LFBFL_i_no_toml=0
549     done
550   fi
551   if [[ LFBFL_i_no_toml -eq 1 ]]; then
552     printf "Running mypy\n"
553     mypy .
554   fi
555   if [[ -n "${LFBFL_mypy_venv}" ]]; then
556     deactivate
557   fi
558
559   printf "Running bandit\n"
560   LFBFL_bandit_venv=""
561   grep_variable "${LFBFL_data_file_path}" bandit_venv\
562     --result-variable-prefix=LFBFL_\
563     --replace-line-returns-by=""
564   if [[ -n "${LFBFL_bandit_venv}" ]]; then
565     maybe_deactivate
566     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
567   fi
568   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
569     pip install --upgrade bandit
570   fi
571   bandit --ini="${LFBFL_subdir3}/bandit.ini"\
572     --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
573     --recursive .
574   # Saving new baseline in temp if necessary.
575   bandit --ini="${LFBFL_subdir3}/bandit.ini"\
576     --format=json\
577     --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
578     --recursive .
579   if [[ -n "${LFBFL_bandit_venv}" ]]; then
580     deactivate
581   fi
582
583   printf "Running pylint\n"
584   LFBFL_pylint_venv=""
585   grep_variable "${LFBFL_data_file_path}" pylint_venv\
586     --result-variable-prefix=LFBFL_\
587     --replace-line-returns-by=""
588   if [[ -n "${LFBFL_pylint_venv}" ]]; then
589     maybe_deactivate
```

```
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_path}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config="${LFBFL_subdir3}/ruff.toml"
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHP native linter\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.php"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      php --syntax-check -f="${LFBFL_file_path}"
    done
  fi

  printf "Running PHP_CodeSniffer\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require squizlabs/php_codesniffer
  fi
  phpcs --report=code\
```

```
636       --standard=build_and_checks_variables/phpcs_ruleset.xml
637
638     printf "Running PHPMD\n"
639     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
640       composer global require phpmd/phpmd
641     fi
642
643     declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
644     local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
645     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
646     readonly LFBFL_temp_phpmd_baseline
647
648     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
649     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
650     readonly LFBFL_phpmd_rulesets
651
652     phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
653       . text "${LFBFL_phpmd_rulesets}"
654     # Saving new baseline in temp if necessary.
655     rm "${LFBFL_temp_phpmd_baseline}"
656     phpmd --color --generate-baseline\
657       --baseline-file="${LFBFL_temp_phpmd_baseline}"\
658       . text "${LFBFL_phpmd_rulesets}"
659     sed --in-place --expression='s/" file="/"\n    file="/g'\
660       "${LFBFL_temp_phpmd_baseline}"
661     sed --in-place --regexp-extended\
662       --expression='s~(file=".*)/>~\1\n  />~g'\
663       "${LFBFL_temp_phpmd_baseline}"
664     diff --suppress-common-lines\
665       "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
666       | grep --invert-match --regexp='<!-- ' --regexp='^[0-9]\+d[0-9]\+$'
667
668     printf "Running phpDocumentor\n"
669     # shellcheck disable=SC2312
670     docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
671     printf -- "---PHP end---\n"
672
673     printf -- "---JS---\n"
674     local LFBFL_npm_lint_directories=""
675     grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
676       --result-variable-prefix=LFBFL_
677     if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
678       printf "Running ESLint\n"
679       local LFBFL_JS_directory
680       declare -a LFBFL_arr_paths
681       mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
```

```
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  printf "Running xmllint\n"
  declare -r LFBFL_s_files_for_xmllint=$(
    find . -type f -iregex ".*\.\(xml\|xhtml\)"\
    | relevant_find\
    | sort
  )
  if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
    declare -a LFBFL_arr_files_for_xmllint
    mapfile -t LFBFL_arr_files_for_xmllint\
      <<< "${LFBFL_s_files_for_xmllint}"
    readonly LFBFL_arr_files_for_xmllint
    local LFBFL_xmllint_files_without_DTD=""
    grep_variable "${LFBFL_data_file_path}" xmllint_files_without_DTD\
      --result-variable-prefix=LFBFL_
    declare -a LFBFL_arr_xmllint_files_without_DTD
    if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
      mapfile -t LFBFL_arr_xmllint_files_without_DTD\
        <<< "${LFBFL_xmllint_files_without_DTD}"
    fi
    for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "xmllint on %s\n" "${LFBFL_file_path}"
      # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
      # xmllint --pedantic --noout -- "${LFBFL_file_path}"
```

```
      xmllint --pedantic --noout "${LFBFL_file_path}"
    done

    local LFBFL_file_path2
    for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
      for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
      do
        if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
          LFBFL_file_path=""
        fi
      done
      if [[ "${LFBFL_file_path}" == "" ]]; then
        continue
      fi
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "xmllint on %s\n" "${LFBFL_file_path}"
      xmllint --pedantic --noout --valid "${LFBFL_file_path}"
    done
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
```

```
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml.
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_s_files_for_Nu=$(
      find . -type f -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
      declare -a LFBFL_arr_files_for_Nu
      mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
      readonly LFBFL_arr_files_for_Nu
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.10 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
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```
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#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap=$(trap -p RETURN)
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
```

```
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  if [[ -z "${LFBFL_previous_return_trap}" ]]; then
    LFBFL_previous_return_trap="trap - RETURN"
  fi
  readonly LFBFL_previous_return_trap
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap2="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap2+="${LFBFL_previous_return_trap}"
  readonly LFBFL_previous_return_trap2
  LFBFL_return_traps_stack+=("${LFBFL_previous_return_trap2}")
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # ------------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ------------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ ${#LFBFL_return_traps_stack[@]} -gt 0 ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    # shellcheck disable=SC2154,SC2309
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ ${#LFBFL_return_traps_stack[@]} -gt 1 ]]; then
        printf "%s executing some return trap: %s.\n"\
```

```
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
      else
        printf "%s setting back to previous_return_trap: %s.\n"\
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
      fi
    fi
    eval -- "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
}


is_top_dirstack_directory(){
  # Altough "${DIRSTACK[0]}" may seem to be already a realpath,
  # possible use of symbolic links makes this function necessary.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
```

```
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd -- "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
```

```
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_some_flag(){
  # $1=variable_name
  # $2=some_flag_long_name
  # $3=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  # This test:
  # if [[ "$*" == *--verbose* ]]; then
  # is faster than below.
  # But it is not bug free in a calling function,
  # and even incompatible with the current function where
```

```
  # the flag_long_name is a mandatory argument.
  printf -v "$1" "%d" 0
  declare -i LFBFL_i
  local LFBFL_argument
  for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
    LFBFL_argument="${!LFBFL_i}"
    if [[ "${LFBFL_argument}" == "$2" ]]; then
      printf -v "$1" "%d" 1
      break
    fi
  done
  if [[ $3 -eq 1 ]]; then
    readonly "$1"
  fi
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  get_some_flag LFBFL_i_verbose --verbose 1 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
  fi
}


get_some_option(){
  # $1=variable_name
  # $2=some_option_long_name
  # $3=option_default_value_if_flag_without_option_value_or_empty_value
  # $4=option_default_value_if_nothing
  # $5=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the four previous.
  # If you truly have a need to distinguish:
  # option_default_value_if_flag_without_option_value --thing
  # option_default_value_if_flag_with_empty_option_value --thing= or =''
  # then think twice before modifying this function.
  printf -v "$1" "%s" "$4"
  declare -i LFBFL_i
  local LFBFL_argument
  for ((LFBFL_i = 6; LFBFL_i <= $#; ++LFBFL_i)); do
```

```
    LFBFL_argument="${!LFBFL_i}"
    if [[ "${LFBFL_argument}" == "$2" ]]; then
      printf -v "$1" "%s" "$3"
      break
    fi
    if [[ "${LFBFL_argument}" == "$2"=* ]]; then
      printf -v "$1" "%s" "${LFBFL_argument#"$2"=}"
      printf -v "$1" "%s" "${!1:-$3}"
      break
    fi
  done
  if [[ $5 -eq 1 ]]; then
    readonly "$1"
  fi
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
# Or:
#   local LFBFL_work_directory=""
```

```
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." "." 0 "$@"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    declare -i LFBFL_i_trap_popd
    get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
    if [[ LFBFL_i_trap_popd -eq 1 ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="popd_from_work_directory 4"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
    fi
  fi
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
```

```
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" "$1" 1 "$@"
}

get_max_line_length_option(){
  # This command is to be called in another one with same arguments.
  # Its most useful feature is that it factorizes the default maximum line
  # length of 70 characters.
  # Options:
  #   --max-line-length
  get_some_option LFBFL_i_max_line_length --max-line-length 70 70 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
```

```
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_unset
  get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
  if [[ LFBFL_i_trap_unset -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
```

```
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_set
  get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
  if [[ LFBFL_i_trap_set -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_unset
  get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
  if [[ LFBFL_i_trap_unset -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
```

```
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_set
  get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
  if [[ LFBFL_i_trap_set -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.11 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_argument}"
      LFBFL_i_first=0
      continue
    fi
```

```
    if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
```

```
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq --repeated
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
```

```
137 }
138
139 max(){
140   # Returns the maximum string among the arguments.
141   # Renvoie la chaîne la plus grande parmi celles proposées.
142   # $1=sort_command with the flags you want to be applied.
143   # $ max sort a b c
144   # c
145   if [[ $# -eq 0 ]]; then
146     return 1
147   fi
148   enhanced_set_shell_option pipefail\
149     && trap 'enhanced_unset_shell_option pipefail' RETURN
150   printf "%s\n" "${@:2}" \
151     | eval -- "$1 --reverse"\
152     | head --lines=1
153 }
154
155 min(){
156   # Returns the minimum string among the arguments.
157   # Renvoie la chaîne la plus petite parmi celles proposées.
158   # $1=sort_command with the flags you want to be applied.
159   # $ min sort a b c
160   # a
161   if [[ $# -eq 0 ]]; then
162     return 1
163   fi
164   enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166   printf "%s\n" "${@:2}" \
167     | eval -- "$1"\
168     | head --lines=1
169 }
170
171 is_substring(){
172   # $1=string
173   # $2=substring
174   declare -r LFBFL_var_1=$(
175     printf "%s" "$1"\
176     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
177   )
178   declare -r LFBFL_var_2=$(
179     printf "%s" "$2"\
180     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
181   )
182   local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
```

```
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php --run="${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php --run="${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.12 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
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# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
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#
# You should have received a copy of
# the GNU General Public License
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```

```
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_path}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_web_url=""
```

```
68    grep_variable "${LFBFL_data_file_path}" repository_web_url\
69      --result-variable-prefix="LFBFL_"\
70      --replace-line-returns-by=""
71
72    local LFBFL_repository_git_url=""
73    grep_variable "${LFBFL_data_file_path}" repository_git_url\
74      --result-variable-prefix="LFBFL_"\
75      --replace-line-returns-by=""
76
77    local LFBFL_abstract=""
78    grep_variable "${LFBFL_data_file_path}" abstract\
79      --result-variable-prefix="LFBFL_"
80
81    local LFBFL_acknowledgments=""
82    grep_variable "${LFBFL_data_file_path}" acknowledgments\
83      --result-variable-prefix="LFBFL_"
84
85    local LFBFL_author_email=""
86    grep_variable "${LFBFL_data_file_path}" author_email\
87      --result-variable-prefix="LFBFL_"\
88      --replace-line-returns-by=""
89
90    local LFBFL_author_full_name=""
91    grep_variable "${LFBFL_data_file_path}" author_full_name\
92      --result-variable-prefix="LFBFL_"\
93      --replace-line-returns-by=" "
94
95    local LFBFL_author_website=""
96    grep_variable "${LFBFL_data_file_path}" author_website\
97      --result-variable-prefix="LFBFL_"\
98      --replace-line-returns-by=""
99
100   declare -i LFBFL_i_max_line_length
101   grep_variable "${LFBFL_data_file_path}" max_line_length\
102     --result-variable-prefix="LFBFL_i_"
103
104   declare -r LFBFL_current_date=$(date --iso-8601="date")
105
106   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
107
108   declare -ir LFBFL_i_number_of_commits=$(
109     git shortlog\
110     | space_starting_lines\
111     | wc --lines
112   )
113
```

```
  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "${LFBFL_common_options[@]}"
    code_lines_count_not_empty "${LFBFL_common_options[@]}"
    code_lines_count_empty "${LFBFL_common_options[@]}"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree_light.txt"

  tree -a -DFh --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree.txt.temp"

  local LFBFL_s_files_paths
  # Remove line returns that are here to keep lines short.
  LFBFL_s_files_paths=$(
    sed --regexp-extended --null-data\
      --expression='s/(.)\\\n/\1/Mg'\
```

```
      -- "${LFBFL_subdir2}/files_names_listing.txt"\
    | grep --invert-match '^//'
  )
  readonly LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  readonly LFBFL_arr_files_paths

  local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
  readonly LFBFL_temp_files_listing
  : > "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties
  LFBFL_i_score_command_properties=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties2
  LFBFL_i_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
```

```
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed --expression='s/\//:/g;s/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
```

```
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed --expression='s/&/\&amp;/g;s/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
    "${LFBFL_temp_files_listing}"
  # Example of generated content of files_listing.tex.sub:
  # ----------------------------------------------------------------------
  # \subsection{
  #   build\_and\_checks\_dependencies/licenses\_templates/%
  # build\_licenses\_templates.exec.sh
  # }
  # \label{
  #   build_and_checks_dependencies:licenses_templates:%
```

```
# build_licenses_templatesexecsh
# }
#
# \VerbatimInput[numbers=left,xleftmargin=-5mm]{
# build_and_checks_dependencies/licenses_templates/%
# build_licenses_templates.exec.sh
# }
# ----------------------------------------------------------------------
sed --in-place --expression="s|\\n</pre>|</pre>|g"\
  -- "${LFBFL_temp_files_listing2}"
overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
  "${LFBFL_temp_files_listing2}"
# Example of generated content of files_listing.html.sub:
# ----------------------------------------------------------------------
# <h3 id="subsection3.15">3.15
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
# </h3>
# ...
# <h3 id="subsection3.33">3.33
# build_and_checks_variables/.gitignore
# </h3>
# <pre class="numbered_lines">
# temp/
# upgrade_venvs_ts
# </pre>
# ----------------------------------------------------------------------
overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
  "${LFBFL_temp_files_lis}"
# Example of generated content of files_lis.html.sub:
# ----------------------------------------------------------------------
#       <li><a href="#subsection3.15">
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
#       </a></li>
# ----------------------------------------------------------------------

# shellcheck disable=SC2248
get_overlength_regexp ${LFBFL_i_max_line_length}

# We verify if some lines are beyond max_line_length characters
# in current_tree_light.txt and current_tree.txt.
declare -ar LFBFL_trees=(
  "current_tree_light.txt"
  "current_tree.txt.temp"
)
```

```
  local LFBFL_tree
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_i_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_i_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(
      grep --regexp="${get_overlength_regexp_result}" --\
        "${LFBFL_tree_path}"
    )
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed --regexp-extended --expression='s/(.*)——[^—]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed --regexp-extended --expression='s/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed --regexp-extended --expression='s|.* ([^ ]+)$|\1|g'
      )
```

```
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed --expression='s/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm --force -- "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" --keep-temp --tree-mode

  local LFBFL_tex_path_start
  LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  readonly LFBFL_tex_path_start
  local LFBFL_html_path_start
  LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
  readonly LFBFL_html_path_start

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
    # Otherwise if there is a tex file, we init the process from it.
```

```
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "${LFBFL_repository_name}.html" ]]; then
    cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed --expression='s/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed --expression='s/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed --in-place\
      --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
      --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
      --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
      --expression="s|@author_email@|${LFBFL_author_email}|g"\
      --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
      --expression="s|@author_website@|${LFBFL_author_website}|g"\
      --expression="s|@current_date@|${LFBFL_current_date}|g"\
      --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
      --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
      --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      -- "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1"\
```

```
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
      "${LFBFL_tex_path_start}.6"

    insert_file_at_token "${LFBFL_tex_path_start}.6"\
      @files_listing_VerbatimInput@\
      "${LFBFL_temp_path}/files_listing.tex.sub"\
      "${LFBFL_tex_path_start}.7"

    overwrite_if_not_equal\
      "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.7" --keep-temp
  fi

  # HTML filling:
  declare -i LFBFL_i_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed --in-place\
      --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
      --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
      --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
      --expression="s|@author_email@|${LFBFL_author_email}|g"\
      --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
      --expression="s|@author_website@|${LFBFL_author_website}|g"\
      --expression="s|@current_date@|${LFBFL_current_date}|g"\
      --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
```

```
      --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
      --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      -- "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1"\
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@\
      "${LFBFL_temp_path}/files_lis.html.sub"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.html.sub"\
      "${LFBFL_html_path_start}.7"

    insert_file_at_token "${LFBFL_html_path_start}.7"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.sub"\
      "${LFBFL_html_path_start}.8"

    overwrite_if_not_equal "${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.8" --keep-temp
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
```

```
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_path}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval -- "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.13 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat -- "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
```

```
50    # $1=base_file_path
51    # $2=target_file_path
52    # $3=line_comment_prefix
53    # $4=optional_post_processing
54    # $5=file_prefix
55    local LFBFL_temp
56    LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57    readonly LFBFL_temp
58    if [[ -n "$5" ]]; then
59      printf "%s\n" "$5"\
60        > "${LFBFL_temp}"
61    fi
62    sed --expression="s/^/$3/g" -- "$1"\
63      >> "${LFBFL_temp}"
64    if [[ -n "$4" ]]; then
65      eval -- "$4"
66    fi
67    overwrite_if_not_equal "$2" "${LFBFL_temp}"
68  }
69
70  split_file_in_two(){
71    # $1=file_path
72    # $2=token assume that token is the only thing on his line,
73    #    except for a possible indentation.
74    # $3=path_of_part1_of_file
75    # $4=path_of_part2_of_file
76    # We cannot add --line-regexp with --fixed-strings because of the
77    # possible indentation in file,
78    # instead we use grep_fixed_string_with_anchor.
79    declare -ir LFBFL_i_line_number=$(
80      grep_fixed_string_with_anchor "$1" "$2"\
81        --enforce-line-ends-with-fixed-string --line-number\
82      | head --lines=1\
83      | cut --delimiter=':' --fields=1
84    )
85    declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
86    declare -ir LFBFL_i_lines_after=$((
87      LFBFL_i_line_count - LFBFL_i_line_number
88    ))
89    head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
90      > "$3"
91    tail --lines="${LFBFL_i_lines_after}" -- "$1"\
92      > "$4"
93  }
94
95  insert_file_at_token(){
```

```
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_file_to_insert
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat -- "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm -- "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "$5"
  fi
}
```

3.14 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
```

```
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.15 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval -- "${LFBFL_eval_string}"
    done
  done
}
```

3.16 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
49    declare -a LFBFL_return_traps_stack
50    init_return_trap
51
52    enhanced_set_shell_option pipefail --trap-unset
53
54    local LFBFL_s_format
55
56    # Source
57    local LFBFL_license_source_subdir
58    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
59    readonly LFBFL_license_source_subdir
60    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
61    LFBFL_source_license_prefix+="license_file_header_"
62    readonly LFBFL_source_license_prefix
63    # Target
64    local LFBFL_license_target_subdir
65    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
66    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
67    readonly LFBFL_license_target_subdir
68    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
69    LFBFL_target_license_prefix+="license_file_header_"
70    readonly LFBFL_target_license_prefix
71
72    declare -ar LFBFL_licenses=(
73      "AGPLv3+"
74      "GPLv3+"
75      "LGPLv3+"
76    )
77
78    # ------------------------------------------------------------------
79    # Various language definitions for text file formats
80    declare -ar LFBFL_block_comment_languages=(
81      "c"
82      "py"
83    )
84    declare -ar LFBFL_block_comment_enters=(
85      '/*'
86      '"""'
87    )
88    declare -ar LFBFL_block_comment_exits=(
89      '*/'
90      '"""'
91    )
92    equal "${#LFBFL_block_comment_languages[@]}"\
93          "${#LFBFL_block_comment_enters[@]}"\
94          "${#LFBFL_block_comment_exits[@]}"
```

```
  declare -i LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
    return 1
  fi

  declare -ar LFBFL_line_comment_languages=(
    "sh"
    "sql"
    "tex"
  )
  declare -ar LFBFL_line_comment_prefixes=(
    '# '
    '-- '
    '% '
  )
  equal "${#LFBFL_line_comment_languages[@]}"\
        "${#LFBFL_line_comment_prefixes[@]}"
  LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_path
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_path
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  declare -i LFBFL_i
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
    LFBFL_license_prefix="${LFBFL_source_license_prefix}"
```

```
141     LFBFL_license_prefix+="${LFBFL_some_license}"
142     LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
143     LFBFL_license_prefix2+="${LFBFL_some_license}"
144     for ((
145       LFBFL_i = 0;
146       LFBFL_i < ${#LFBFL_block_comment_languages[@]};
147       ++LFBFL_i
148     )); do
149       LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
150       LFBFL_license_file_path="${LFBFL_license_prefix2}"
151       LFBFL_license_file_path+=".${LFBFL_extension}"
152       LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
153       LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
154       generate_from_template_with_block_comments\
155         "${LFBFL_license_prefix}.tpl"\
156         "${LFBFL_license_file_path}.tpl"\
157         "${LFBFL_enter_string}" "${LFBFL_exit_string}"
158       LFBFL_i_generate_from_template_result=$?
159       if [[ LFBFL_i_verbose -eq 1 ]]; then
160         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
161           printf "License template %s.tpl updated.\n"\
162             "${LFBFL_license_file_path}"
163         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
164           printf "License template %s.tpl created.\n"\
165             "${LFBFL_license_file_path}"
166         fi
167       fi
168     done
169     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
170     for ((
171       LFBFL_i = 0;
172       LFBFL_i < ${#LFBFL_line_comment_languages[@]};
173       ++LFBFL_i
174     )); do
175       LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
176       LFBFL_license_file_path="${LFBFL_license_prefix2}"
177       LFBFL_license_file_path+=".${LFBFL_extension}"
178       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
179       LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
180       LFBFL_intermediate_file_path+="${LFBFL_temp2}"
181       LFBFL_file_prefix=""
182       if [[ "${LFBFL_extension}" == "sh" ]]; then
183         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
184       fi
185       # The sed command is here to remove trailing spaces of previously
186       # empty lines after comment prefix is added to the line.
```

```
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_path}'"\
        "${LFBFL_file_prefix}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_path="build_and_checks_variables/"
  LFBFL_data_file_path+="repository_data.txt"
  readonly LFBFL_data_file_path
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_path}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_path}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_path}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_path}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
  readonly LFBFL_license_prefix
```

```
233   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
234   readonly LFBFL_license_prefix2
235   local LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
236   readonly LFBFL_license_prefix3
237
238   # First year according to current state of git repository.
239   declare -ir LFBFL_i_first_year=$(
240     git log\
241     | grep 'Date:'\
242     | tail --lines=1\
243     | cut --delimiter=' ' --fields=8
244   )
245   # Last year according to current state of git repository.
246   declare -ir LFBFL_i_last_year=$(
247     git log\
248     | grep 'Date:'\
249     | head --lines=1\
250     | cut --delimiter=' ' --fields=8
251   )
252   local LFBFL_copyright_string
253   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
254   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
255   readonly LFBFL_copyright_string
256
257   prepare_filled_license_file_block(){
258     # $1=license_file_path
259     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
260         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
261         -- "$1.tpl"\
262       > "$1.temp"
263     overwrite_if_not_equal "$1" "$1.temp"
264     declare -ir LFBFL_i_result=$?
265     if [[ LFBFL_i_verbose -eq 1 ]]; then
266       if [[ LFBFL_i_result -eq 1 ]]; then
267         printf "License template %s updated.\n" "$1"
268       elif [[ LFBFL_i_result -eq 2 ]]; then
269         printf "License template %s created.\n" "$1"
270       fi
271     fi
272     head --lines=-1 -- "$1"\
273       | tail --lines=+2\
274       > "$1.temp"
275     # Strange world: tail --lines=-1 was working
276     # to exclude first line.
277     # I tested it months ago, and saw with my eyes the start and end
278     # block comments removed with:
```

```
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
    ["js"]="c"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_path2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_path}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_path2="${LFBFL_license_prefix3}"
      LFBFL_license_file_path2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_path2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
        "${LFBFL_license_file_path}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
          "${LFBFL_license_file_path2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm -- "${LFBFL_license_file_path}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm -- "${LFBFL_license_file_path2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_path
    sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
        --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
        -- "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf "License template %s created.\n" "$1"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["sql"]="sql"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
```

```
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_path}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_path2="${LFBFL_license_prefix3}"
      LFBFL_license_file_path2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_path2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

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```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

```
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```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```
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```

3.20 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
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```

```
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#
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "all_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep --invert-match --perl-regexp --with-filename\
        -- 'a(?!a)a'
  done
}
```

```
62
63 all_self_code_lines(){
64   # Options:
65   #   --verbose
66   #   --work-directory=""
67   declare -i LFBFL_i_verbose=0
68   get_verbose_option "$@"
69   local LFBFL_work_directory=""
70   get_work_directory_option "$@"
71   declare -a LFBFL_return_traps_stack
72   init_return_trap
73   pushd_to_work_directory --trap-popd
74   can_continue_after_enhanced_pushd || return 1
75
76   enhanced_set_shell_option pipefail --trap-unset
77
78   declare -a LFBFL_common_options=("$@")
79   keep_options LFBFL_common_options --verbose
80
81   # below and another lines, see SC2145 false positive
82   all_code_lines "${LFBFL_common_options[@]}" \
83     | relevant_grep\
84     | not_license_grep\
85     | not_main_tex_grep\
86     | not_main_html_grep
87 }
88
89 # shellcheck disable=SC2120
90 all_self_empty_code_lines(){
91   # Options:
92   #   --verbose
93   #   --work-directory=""
94   declare -i LFBFL_i_verbose=0
95   get_verbose_option "$@"
96   local LFBFL_work_directory=""
97   get_work_directory_option "$@"
98   declare -a LFBFL_return_traps_stack
99   init_return_trap
100  pushd_to_work_directory --trap-popd
101  can_continue_after_enhanced_pushd || return 1
102
103  enhanced_set_shell_option pipefail --trap-unset
104
105  declare -a LFBFL_common_options=("$@")
106  keep_options LFBFL_common_options --verbose
107
```

```
  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_all_result
  i_code_lines_count_all_result=$(
    all_self_code_lines\
    | wc --lines
```

```
154   )
155   # The printf is for command line use.
156   # A script should use the result value instead.
157   # shellcheck disable=SC2248
158   printf "%s\n" ${i_code_lines_count_all_result}
159 }
160
161 code_lines_count_empty(){
162   # Options:
163   #   --verbose
164   #   --work-directory=""
165   declare -i LFBFL_i_verbose=0
166   get_verbose_option "$@"
167   local LFBFL_work_directory=""
168   get_work_directory_option "$@"
169   declare -a LFBFL_return_traps_stack
170   init_return_trap
171   pushd_to_work_directory --trap-popd
172   can_continue_after_enhanced_pushd || return 1
173
174   enhanced_set_shell_option pipefail --trap-unset
175
176   declare -gi i_code_lines_count_empty_result
177   # shellcheck disable=SC2119
178   i_code_lines_count_empty_result=$(
179     all_self_empty_code_lines\
180     | wc --lines
181   )
182   # The printf is for command line use.
183   # A script should use the result value instead.
184   # shellcheck disable=SC2248
185   printf "%s\n" ${i_code_lines_count_empty_result}
186 }
187
188 code_lines_count_not_empty(){
189   # Options:
190   #   --verbose
191   #   --work-directory=""
192   declare -i LFBFL_i_verbose=0
193   get_verbose_option "$@"
194   local LFBFL_work_directory=""
195   get_work_directory_option "$@"
196   declare -a LFBFL_return_traps_stack
197   init_return_trap
198   pushd_to_work_directory --trap-popd
199   can_continue_after_enhanced_pushd || return 1
```

```

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.21 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
```

```
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
```

```
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat -- "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
```

```
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
    rm -- "${LFBFL_temp}"
    return
  fi
  mv -- "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below,
  #   you don't need to supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
```

```
    ]]; then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[
      "${LFBFL_argument}" == "-q"
      || "${LFBFL_argument}" == "--quiet"
    ]]; then
      LFBFL_i_quiet=1
      LFBFL_grep_options+=("${LFBFL_argument}")
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_argument}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
  ]]; then
    grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
    return
  fi
  declare -r LFBFL_s_lines=$(
    grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
      "$2" "$1"
  )
  declare -i LFBFL_i_result=1
  if [[ -z "${LFBFL_s_lines}" ]]; then
    # shellcheck disable=SC2248
    return ${LFBFL_i_result}
  fi
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  declare -i LFBFL_i_fixed_string_length=${#2}
  local LFBFL_some_line
  local LFBFL_line_part
  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "$2" ]]; then
      LFBFL_i_result=0
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        break
      else
        printf "%s\n" "${LFBFL_some_line}"
      fi
```

```
      continue
    fi
    if [[
      LFBFL_i_enforce_line_starts -eq 1
      && LFBFL_i_enforce_line_ends -eq 1
    ]]; then
      continue
    fi
    if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
    fi
    if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
    fi
    if [[ "${LFBFL_line_part}" == "$2" ]]; then
      LFBFL_i_result=0
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        break
      else
        printf "%s\n" "${LFBFL_some_line}"
      fi
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_result}
}

grep_variable(){
  # $1=file_path
  # $2=variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  # printf "%s\n" "${LFBFL_regexp}"
  local LFBFL_variable_value
  local LFBFL_s_replace
  get_some_option LFBFL_s_replace --replace-line-returns-by '' $'\n' 1 "$@"
  if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
    # sed -Ez -e...
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"\
      | sed --null-data --regexp-extended\
        --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
```

```
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n" "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep --extended-regexp --invert-match '^( |$)'
}
```

```
all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep --invert-match '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep --invert-match '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep --invert-match '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep --extended-regexp '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
```

```
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_php_cache_find(){
```

```
  grep --extended-regexp --invert-match\
    --regexp="(^|/)\.phpdoc/cache/"
}

not_php_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?\.phpdoc/cache/"
}

not_python_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)__pycache__/"\
    --regexp="(^|/)\.mypy_cache/"\
    --regexp="(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_php_cache_find\
    | not_python_cache_find
}

not_cache_grep(){
  not_php_cache_grep\
    | not_python_cache_grep
}

not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
```

```
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}
```

```
490 not_main_html_grep(){
491   local LFBFL_repository_name=""
492   grep_variable build_and_checks_variables/repository_data.txt\
493     repository_name\
494     --result-variable-prefix="LFBFL_"\
495     --replace-line-returns-by=""
496   grep --extended-regexp --invert-match\
497     --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
498 }
499
500 not_temp_file_find(){
501   grep --extended-regexp --invert-match\
502     --regexp="(^|/)build_and_checks_variables/temp/"
503 }
504
505 not_temp_file_grep(){
506   grep --extended-regexp --invert-match\
507     --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
508 }
509
510 not_unchecked_builds_find(){
511   grep --extended-regexp --invert-match\
512     --regexp="(^|/)\.phpdoc/build/"
513 }
514
515 not_unchecked_builds_grep(){
516   grep --extended-regexp --invert-match\
517     --regexp="^([^:]+/)?\.phpdoc/build/"
518 }
519
520 relevant_find(){
521   enhanced_set_shell_option pipefail\
522     && trap 'enhanced_unset_shell_option pipefail' RETURN
523   not_dependencies_find\
524     | not_cache_find\
525     | not_git_find\
526     | not_archive_find\
527     | not_temp_file_find\
528     | not_unchecked_builds_find
529 }
530
531 relevant_grep(){
532   enhanced_set_shell_option pipefail\
533     && trap 'enhanced_unset_shell_option pipefail' RETURN
534   not_dependencies_grep\
535     | not_cache_grep\
```

```
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep\
    | not_unchecked_builds_grep
}
```

3.22 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed\
```

```
39      --expression='s/\xc2\xa0/ /g'\
40      --expression='s/\xe2\x80\x80/ /g'\
41      --expression='s/\xe2\x80\x81/ /g'\
42      --expression='s/\xe2\x80\x82/ /g'\
43      --expression='s/\xe2\x80\x83/ /g'\
44      --expression='s/\xe2\x80\x84/ /g'\
45      --expression='s/\xe2\x80\x85/ /g'\
46      --expression='s/\xe2\x80\x86/ /g'\
47      --expression='s/\xe2\x80\x87/ /g'\
48      --expression='s/\xe2\x80\x88/ /g'\
49      --expression='s/\xe2\x80\x89/ /g'\
50      --expression='s/\xe2\x80\x8a/ /g'\
51      --expression='s/\xe2\x80\x8b/ /g'\
52      --expression='s/\xe2\x80\x8c/ /g'\
53      --expression='s/\xe2\x80\x8d/ /g'\
54      --expression='s/\xe2\x80\x8e/ /g'\
55      --expression='s/\xe2\x80\x8f/ /g'\
56      --expression='s/\xe2\x80\xa8/ /g'\
57      --expression='s/\xe2\x80\xa9/ /g'\
58      --expression='s/\xe2\x80\xaa/ /g'\
59      --expression='s/\xe2\x80\xab/ /g'\
60      --expression='s/\xe2\x80\xac/ /g'\
61      --expression='s/\xe2\x80\xad/ /g'\
62      --expression='s/\xe2\x80\xae/ /g'\
63      --expression='s/\xe2\x80\xaf/ /g'\
64      --expression='s/\xe2\x81\x9f/ /g'\
65      --expression='s/\xe2\x81\xa0/ /g'\
66      --expression='s/\xe2\x81\xa1/ /g'\
67      --expression='s/\xe2\x81\xa2/ /g'\
68      --expression='s/\xe2\x81\xa3/ /g'\
69      --expression='s/\xe2\x81\xa4/ /g'\
70      --expression='s/\xe2\x81\xa5/ /g'\
71      --expression='s/\xe2\x81\xa6/ /g'\
72      --expression='s/\xe2\x81\xa7/ /g'\
73      --expression='s/\xe2\x81\xa8/ /g'\
74      --expression='s/\xe2\x81\xa9/ /g'\
75      --expression='s/\xe2\x81\xaa/ /g'\
76      --expression='s/\xe2\x81\xab/ /g'\
77      --expression='s/\xe2\x81\xac/ /g'\
78      --expression='s/\xe2\x81\xad/ /g'\
79      --expression='s/\xe2\x81\xae/ /g'\
80      --expression='s/\xe2\x81\xaf/ /g'
81  }
```

3.23 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # Options:
  #   --keep-temp : keep the temporary file
  #   --quiet : we even don't want the message of diff saying that
  #             files are distinct
  #   --tree-mode : when the two files are the output of tree command,
  #   we ignore if all the differences are only on directories timestamps.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  declare -i LFBFL_i_keep_temp
  get_some_flag LFBFL_i_keep_temp --keep-temp 1 "$@"
```

```
45    declare -i LFBFL_i_quiet
46    get_some_flag LFBFL_i_quiet --quiet 1 "$@"
47    declare -i LFBFL_i_tree_mode
48    get_some_flag LFBFL_i_tree_mode --tree-mode 1 "$@"
49
50    if [[ ! -f "$1" ]]; then
51      if [[ LFBFL_i_keep_temp -eq 1 ]]; then
52        cp --preserve=mode,ownership,timestamps -- "$2" "$1"
53      else
54        mv -- "$2" "$1"
55      fi
56      return 2
57    fi
58    declare -i LFBFL_i_is_equal
59    if [[ LFBFL_i_tree_mode -eq 1 ]]; then
60      # Not setting pipefail since the result would still be incorrect.
61      # shellcheck disable=SC2312
62      diff -- "$1" "$2"\
63        | grep --extended-regexp "^(>|<)"\
64        | grep --invert-match "/$"
65      LFBFL_i_is_equal=${PIPESTATUS[2]}
66      if [[ LFBFL_i_is_equal -eq 1 ]]; then
67        grep --only-matching --regexp="[^ ]*/$" -- "$1"\
68          > "overwrite_if_not_equal.file1.temp"
69        grep --only-matching --regexp="[^ ]*/$" -- "$2"\
70          > "overwrite_if_not_equal.file2.temp"
71        if [[ LFBFL_i_quiet -eq 1 ]]; then
72          diff overwrite_if_not_equal.file1.temp\
73            overwrite_if_not_equal.file2.temp\
74            > /dev/null
75        else
76          diff overwrite_if_not_equal.file1.temp\
77            overwrite_if_not_equal.file2.temp
78        fi
79        LFBFL_i_is_equal=1-$?
80        rm overwrite_if_not_equal.file1.temp\
81          overwrite_if_not_equal.file2.temp
82      fi
83    else
84      if [[ LFBFL_i_quiet -eq 1 ]]; then
85        diff --brief -- "$1" "$2" > /dev/null
86      else
87        diff --brief -- "$1" "$2"
88      fi
89      LFBFL_i_is_equal=1-$?
90    fi
```

```
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ LFBFL_i_keep_temp -eq 0 ]]; then
      rm -- "$2"
    fi
    return 0
  fi
  if [[ LFBFL_i_keep_temp -eq 1 ]]; then
    cp --preserve=mode,ownership,timestamps -- "$2" "$1"
  else
    mv -- "$2" "$1"
  fi
  return 1
}
```

3.24 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
```

```
31 # https://github.com/psf/black/issues/2370
32 # And moreover, at some point, I saw the reverse rule
33 # in black or pylint to add an empty line...
34 # This regexp may give false positives,
35 # but that's not the end of the world.
36 check_no_empty_line_after_python_function_docstring(){
37   # Options:
38   #   --verbose
39   #   --work-directory=""
40   declare -i LFBFL_i_verbose=0
41   get_verbose_option "$@"
42   local LFBFL_work_directory=""
43   get_work_directory_option "$@"
44   declare -a LFBFL_return_traps_stack
45   init_return_trap
46   pushd_to_work_directory --trap-popd
47   can_continue_after_enhanced_pushd || return 1
48
49   enhanced_set_bash_option globstar --trap-unset
50
51   [[ LFBFL_i_verbose -eq 1 ]]\
52     && printf "Checking empty lines after Python function docstrings\n"
53   pcre2grep --multiline --\
54     $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
55     **/*.py
56 }
57
58 python_black_complement(){
59   # Options:
60   #   --verbose
61   #   --work-directory=""
62   declare -i LFBFL_i_verbose=0
63   get_verbose_option "$@"
64   local LFBFL_work_directory=""
65   get_work_directory_option "$@"
66   declare -a LFBFL_return_traps_stack
67   init_return_trap
68   pushd_to_work_directory --trap-popd
69   can_continue_after_enhanced_pushd || return 1
70
71   declare -a LFBFL_common_options=("$@")
72   keep_options LFBFL_common_options --verbose
73
74   check_no_empty_line_after_python_function_docstring\
75     "${LFBFL_common_options[@]}"
76 }
```

3.25 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # This function should be useless in a few months:
  # https://github.com/PyCQA/isort/issues/2250
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
45    declare -a LFBFL_return_traps_stack
46    init_return_trap
47    pushd_to_work_directory --trap-popd
48    can_continue_after_enhanced_pushd || return 1
49
50    [[ LFBFL_i_verbose -eq 1 ]]\
51      && printf\
52        "Checking that import of _collections_abc is at the right place\n"
53
54    enhanced_set_shell_option pipefail --trap-unset
55
56    declare -r LFBFL_temp=".check_collections_abc_place.temp"
57    local LFBFL_file_path
58    declare -r LFBFL_s_files_paths=$(
59      find . -type f -name "*.py" -printf '%P\n'\
60      | relevant_find
61    )
62    if [[ -z "${LFBFL_s_files_paths}" ]]; then
63      [[ LFBFL_i_verbose -eq 1 ]]\
64        && printf "check_collections_abc_place: No file found.\n"
65      return
66    fi
67    declare -a LFBFL_arr_files_paths
68    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
69    readonly LFBFL_arr_files_paths
70    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
71      sed --regexp-extended --null-data\
72        --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
73        -- "${LFBFL_file_path}"\
74        > "${LFBFL_file_path}${LFBFL_temp}"
75      overwrite_if_not_equal "${LFBFL_file_path}"\
76        "${LFBFL_file_path}${LFBFL_temp}"
77    done
78  }
79
80  python_isort_complement(){
81    # Options:
82    #   --verbose
83    #   --work-directory=""
84    declare -i LFBFL_i_verbose=0
85    get_verbose_option "$@"
86    local LFBFL_work_directory=""
87    get_work_directory_option "$@"
88    declare -a LFBFL_return_traps_stack
89    init_return_trap
90    pushd_to_work_directory --trap-popd
```

```
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.26 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
```

```
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

3.27 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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```

```
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#
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion.\n"
  # shellcheck disable=SC2059
  [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"

  pcre2grep --multiline -- $'\\$\\{#(|.*_)arr_(?!.*\\[@\\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
```

```
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];[^\n]{0,${LFBFL_i_max_previous_line_length}})"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"
  # pcre2grep --multiline --
  # '^[^\n]{1,70}(?<=\]\];[^\n]{0,70})'
  # '(?<!^\s{0,70}\]\];)\n\s*then'
  pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh
}

check_no_negation_before_bash_test(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking that shell scripts do not contain ! before [[.\n"

  pcre2grep --multiline -- $'!\\s*\\[\\[' **/*.sh
}
```

```

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

3.28 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

/**
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```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once './build_and_checks_dependencies/split_score.libr.php';

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
// phpcs:enable Generic.WhiteSpace.ArbitraryParenthesesSpacing
```

```
?>
```

3.29 build_and_checks_dependencies/split_score.libr.php

```
<?php

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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

```

```
43
44
45 /**
46 Given the 3 following arguments, this function returns a closure that can
47 be used to compute the split score given the current string
48 (hopefully matching a delimiter), the current cut/split position,
49 and if that split position is after or before the current string.
50
51 @param bool          $b_larger_after
52                      Score is larger when splitting after,
53                      previous name of this parameter was $b_after_before.
54 @param int           $i_max_length
55                      The maximum length allowed (goal length).
56 @param array<string> $a_delimiter_strings_domain
57                      An array of "delimiters".
58
59 @return \Closure {
60   The "split_score" closure to use repeatedly afterward.
61
62   @param string $s_delimiter_string The string tested to be a delimiter.
63   @param int    $i_cut_position     The offset of the cut in the string.
64   @param bool   $b_is_cut_after     Is the offset taken after the string?
65
66   @return int The split score.
67 }
68 */
69 function generate_split_score(
70   bool $b_larger_after,
71   int $i_max_length,
72   array $a_delimiter_strings_domain,
73 ) : Closure {
74   return function (
75     string $s_delimiter_string,
76     int $i_cut_position,
77     bool $b_is_cut_after,
78   ) use (
79     $b_larger_after,
80     $i_max_length,
81     $a_delimiter_strings_domain,
82   ) : int {
83     if(
84       !in_array(
85         $s_delimiter_string,
86         $a_delimiter_strings_domain,
87         true,
88       )
```

```
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_larger_after){
      // When $b_larger_after,
      // always way better to cut after '/' than before it.
      // When !$b_larger_after (aka larger_before),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}//end generate_split_score()
?>
```

3.30 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
```

```
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-----------------------------------------------------------------------
#-----------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-----------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
```

```
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
```

```
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
```

```
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 - 1 and RP7 : p1 = p2 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that’s not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
```

```
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
```

```
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
```

```
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
```

```
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
```

```
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

```

```
# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
```

```
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
```

```
532       | SSP_POSITION_NOT_DECREASING
533       | SSP_POSITION_NOT_DECREASING_AFTER
534       | SSP_POSITION_NOT_DECREASING_BEFORE
535       | SSP_POSITION_UNIFORM
536       | SSP_POSITION_UNIFORM_AFTER
537       | SSP_POSITION_UNIFORM_BEFORE
538       | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
539       | SSP_SAME_DELIMITER_LARGER_AFTER
540       | SSP_SAME_DELIMITER_LARGER_BEFORE
541       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
542       | SSP_SAME_POSITION_LARGER_AFTER
543       | SSP_SAME_POSITION_LARGER_BEFORE
544       | SSP_UNIFORM
545     ))
546     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
547       LFBFL_i_result_old=$((LFBFL_i_result))
548       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
549         LFBFL_i_result=$((
550           LFBFL_i_result
551           | SSP_LARGER_AFTER
552           | SSP_SAME_DELIMITER_LARGER_AFTER
553           | SSP_SAME_POSITION_LARGER_AFTER
554         ))
555       fi
556       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558           LFBFL_i_result
559           | SSP_LARGER_BEFORE
560           | SSP_SAME_DELIMITER_LARGER_BEFORE
561           | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563       fi
564       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566           LFBFL_i_result
567           | SSP_POSITION_NOT_DECREASING_AFTER
568           | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570       fi
571       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((
573           LFBFL_i_result
574           | SSP_POSITION_UNIFORM_AFTER
575           | SSP_POSITION_UNIFORM_BEFORE
576           | SSP_POSITION_NOT_DECREASING
577           | SSP_POSITION_NOT_DECREASING_AFTER
```

```
578         | SSP_POSITION_NOT_DECREASING_BEFORE
579       ))
580     fi
581     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582       LFBFL_i_result=$((
583         LFBFL_i_result
584         | SSP_POSITION_NOT_DECREASING_AFTER
585       ))
586     fi
587     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588       LFBFL_i_result=$((
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591       ))
592     fi
593     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594       LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601       ))
602     fi
603     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604       LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607       ))
608     fi
609     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610       LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613       ))
614     fi
615     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616       LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM
619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
```

```
624         ))
625       fi
626       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627         LFBFL_i_result=$((
628           LFBFL_i_result
629           | SSP_LARGER_AFTER
630         ))
631       fi
632       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633         LFBFL_i_result=$((
634           LFBFL_i_result
635           | SSP_LARGER_BEFORE
636         ))
637       fi
638       if ((LFBFL_i_result & SSP_UNIFORM)); then
639         LFBFL_i_result=$((LFBFL_i_all_flags))
640         break
641       fi
642       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643       && ((
644         LFBFL_i_result & (
645           SSP_ALWAYS_LARGER_BEFORE
646           | SSP_EQUAL_AFTER_OR_BEFORE
647           | SSP_LARGER_BEFORE
648           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649           | SSP_SAME_DELIMITER_LARGER_BEFORE
650           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651           | SSP_SAME_POSITION_LARGER_BEFORE
652         )
653       )); then
654         LFBFL_i_result=$((LFBFL_i_all_flags))
655         break
656       fi
657       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658       && ((
659         LFBFL_i_result & (
660           SSP_EQUAL_AFTER_OR_BEFORE
661           | SSP_LARGER_AFTER
662           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663           | SSP_SAME_DELIMITER_LARGER_AFTER
664           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
665           | SSP_SAME_POSITION_LARGER_AFTER
666         )
667       )); then
668         LFBFL_i_result=$((LFBFL_i_all_flags))
669         break
```

```
670     fi
671     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
672     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
673       LFBFL_i_result=$((
674         LFBFL_i_result
675         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
676       ))
677     fi
678     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
679     && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
680       LFBFL_i_result=$((
681         LFBFL_i_result
682         | SSP_SAME_POSITION_LARGER_AFTER
683       ))
684     fi
685     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
686     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
687       LFBFL_i_result=$((
688         LFBFL_i_result
689         | SSP_SAME_POSITION_LARGER_BEFORE
690       ))
691     fi
692     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
693     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
694       LFBFL_i_result=$((LFBFL_i_all_flags))
695       break
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
699       LFBFL_i_result=$((
700         LFBFL_i_result
701         | SSP_ALWAYS_LARGER_AFTER
702       ))
703     fi
704     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
705     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
706       LFBFL_i_result=$((
707         LFBFL_i_result
708         | SSP_ALWAYS_LARGER_BEFORE
709       ))
710     fi
711     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
712     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
713       LFBFL_i_result=$((
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
```

```
716         ))
717       fi
718       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719       && ((
720         LFBFL_i_result & (
721           SSP_POSITION_UNIFORM
722           | SSP_POSITION_UNIFORM_AFTER
723         )
724       )); then
725         LFBFL_i_result=$((
726           LFBFL_i_result
727           | SSP_SAME_DELIMITER_LARGER_AFTER
728         ))
729       fi
730       if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731       && ((
732         LFBFL_i_result & (
733           SSP_POSITION_UNIFORM
734           | SSP_POSITION_UNIFORM_BEFORE
735         )
736       )); then
737         LFBFL_i_result=$((
738           LFBFL_i_result
739           | SSP_SAME_DELIMITER_LARGER_BEFORE
740         ))
741       fi
742       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743       && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744         LFBFL_i_result=$((
745           LFBFL_i_result
746           | SSP_POSITION_NOT_DECREASING
747         ))
748       fi
749       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751         LFBFL_i_result=$((
752           LFBFL_i_result
753           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754         ))
755       fi
756       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
757       && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758         LFBFL_i_result=$((LFBFL_i_all_flags))
759         break
760       fi
761       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
```

```
762     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763       LFBFL_i_result=$((
764         LFBFL_i_result
765         | SSP_ALWAYS_LARGER_AFTER
766       ))
767     fi
768     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770       LFBFL_i_result=$((
771         LFBFL_i_result
772         | SSP_ALWAYS_LARGER_BEFORE
773       ))
774     fi
775     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776     && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777       LFBFL_i_result=$((
778         LFBFL_i_result
779         | SSP_POSITION_UNIFORM
780       ))
781     fi
782     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784       LFBFL_i_result=$((
785         LFBFL_i_result
786         | SSP_ALWAYS_LARGER_AFTER
787       ))
788     fi
789     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791       LFBFL_i_result=$((
792         LFBFL_i_result
793         | SSP_ALWAYS_LARGER_BEFORE
794       ))
795     fi
796     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797     && ((
798       LFBFL_i_result & (
799         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800         | SSP_SAME_POSITION_LARGER_AFTER
801         | SSP_SAME_POSITION_LARGER_BEFORE
802       )
803     )); then
804       LFBFL_i_result=$((LFBFL_i_all_flags))
805       break
806     fi
807     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
```

```
808       && ((
809         LFBFL_i_result & (
810           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811           | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813       )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816       fi
817       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819         LFBFL_i_result=$((
820           LFBFL_i_result
821           | SSP_ALWAYS_LARGER_AFTER
822         ))
823       fi
824       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825       && ((
826         LFBFL_i_result & (
827           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828           |  SSP_SAME_POSITION_LARGER_AFTER
829         )
830       )); then
831         LFBFL_i_result=$((LFBFL_i_all_flags))
832         break
833       fi
834       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
835       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
836         LFBFL_i_result=$((
837           LFBFL_i_result
838           | SSP_ALWAYS_LARGER_BEFORE
839         ))
840       fi
841       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
842       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
843         LFBFL_i_result=$((
844           LFBFL_i_result
845           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
846         ))
847       fi
848     done
849     declare -gi i_split_score_properties_logical_closure_result=$((
850       LFBFL_i_result
851     ))
852   }
853
```

```
split_score_simple(){
  # $1=larger_after (The name was "after_before" previously,
  #                  the meaning was not modified.
  #                  Remark not repeated below.)
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
    i_split_score_result=2
    return
  fi
  i_split_score_result=1
}

get_split_score_simple(){
  # $1=larger_after
  # $2=max_length
  # $3=delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   i_get_split_score_result2=7
  # else
  #   i_get_split_score_result2=11
  # fi
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_POSITION_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
```

```
900   ))
901 }
902
903 get_split_score_exec(){
904   declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
905   SPLIT_SCORE_EXEC+="/split_score.exec.php"
906 }
907
908 split_score(){
909   # $1=larger_after
910   # $2=max_length
911   # $3=delimiters_strings_domain concatenated characters/delimiters
912   # $4=delimiter_string a single character
913   # $5=cut_position
914   # $6=is_cut_after
915   get_split_score_exec
916   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
917   declare -gi i_split_score_result
918   i_split_score_result=$(eval -- "${LFBFL_command}")
919 }
920
921 # This function is currently not called,
922 # hence we need this shellcheck disable.
923 # shellcheck disable=SC2034
924 get_split_score(){
925   # $1=larger_after
926   # $1=max_length
927   # $2=delimiters_strings_domain
928   declare -g get_split_score_result="split_score $1 $2 '$3'"
929   declare -gi i_get_split_score_result2
930   # i_get_split_score_result2=$((9-$1*4))
931   i_get_split_score_result2=$((
932     SSP_DELIMITER_UNIFORM
933     + SSP_LARGER_BEFORE
934     + (
935       $1 * (
936         SSP_LARGER_AFTER
937         - SSP_LARGER_BEFORE
938       )
939     )
940   ))
941   # shellcheck disable=SC2248
942   split_score_properties_logical_closure ${i_get_split_score_result2}
943   i_get_split_score_result2=$((
944     i_split_score_properties_logical_closure_result
945   ))
```

```
}

split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=forbidden_previous_character
  # $6=split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
  #
  # Note that we're always looking for the last position to split
  # among the positions with maximum score;
  # if some position gets a score from 2 delimiters,
  # its score is always the maximum of the two corresponding scores.
  declare -g split_line_at_most_result_start
  declare -g split_line_at_most_result_end
  declare -A LFBFL_positions
  # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
  # For my use case in Bash scripts, I will need only an array of
  # characters. See get_split_score().
  if [[ -n "$6" ]]; then
```

```
    printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
  fi
  if [[ $2 -ge ${#1} ]]; then
    split_line_at_most_result_start=$1
    split_line_at_most_result_end=""
    return
  fi
  if [[ $2 -lt 1 ]]; then
    printf "split_line_at_most: max_length should be at least 1...\n"
    return
  fi
  # We know that ${#1} > $2, hence ${#1} - 1 >= $2
  # Edge cases:
  # There is always at least one character that is overlength,
  # but we need to look only at the score before that character that is
  # overlength, the score after that character is invalid,
  # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
  # That first character that is overlength has index $2.
  # The cut position after that character has index $2 + 1.
  # Looping from character position going from 0 to $2, we need to drop
  # score before character 0 and drop score after character $2.
  # We do this with forbidden score -1.
  declare -ir LFBFL_i_max=$(($2))
  declare -i LFBFL_i
  declare -i LFBFL_j
  local LFBFL_current_char

  # At the beginning, each position has score 0.
  LFBFL_positions["0"]="-1"
  for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
    LFBFL_positions["${LFBFL_i}"]="0"
  done
  LFBFL_positions["${LFBFL_i}"]="-1"

  # If we have a forbidden previous character,
  # we mark the forbidden positions with -1.
  if [[ -n "$5" ]]; then
    for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      ((++LFBFL_i))
      if [[ "${LFBFL_current_char}" == "$5" ]]; then
        LFBFL_positions["${LFBFL_i}"]="-1"
      fi
    done
  fi

```

```
1038    declare -r LFBFL_sort_command="sort --numeric-sort"
1039
1040    update_position_score(){
1041      # $1=position
1042      # Returns 0 if updated, 1 otherwise
1043      # printf "%s|%s\n" "${i_split_score_result}" "$1"
1044      if [[ i_split_score_result -ge 1 ]]; then
1045        # printf "valid result"
1046        if [[ ${LFBFL_positions["$1"]} != "-1" ]]; then
1047          LFBFL_positions["$1"]=$(
1048            max "${LFBFL_sort_command}"\
1049              "${LFBFL_positions["$1"]}"\
1050              "${i_split_score_result}"
1051          )
1052          return 0
1053        fi
1054      fi
1055      return 1
1056    }
1057
1058    local LFBFL_split_score_command=$3
1059    declare -i LFBFL_split_score_command_properties=$4
1060
1061    normal_or_position_uniform_call(){
1062      # $1=cut_position
1063      # $2=is_cut_after
1064      if ((LFBFL_split_score_command_properties & SSP_POSITION_UNIFORM));
1065      then
1066        eval --\
1067          "${LFBFL_split_score_command} '${LFBFL_current_char}' $2"
1068      else
1069        eval --\
1070          "${LFBFL_split_score_command} '${LFBFL_current_char}' $1 $2"
1071      fi
1072    }
1073
1074    if (($4 & SSP_DELIMITER_UNIFORM))\
1075    && (($4 & SSP_LARGER_AFTER))\
1076    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1077      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1078        LFBFL_current_char="${1:${LFBFL_i}:1}"
1079        LFBFL_j=$((LFBFL_i + 1))
1080        # shellcheck disable=SC2248
1081        normal_or_position_uniform_call ${LFBFL_j} 1
1082        # shellcheck disable=SC2248
1083        if update_position_score ${LFBFL_j}
```

```
      then
        break
      fi

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
    done
  elif (($4 & SSP_DELIMITER_UNIFORM))\
  && (($4 & SSP_LARGER_BEFORE))\
  && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      # We still need to break in second if, after computing score after,
      # because SSP_LARGER_BEFORE implies large inequality instead of
      # strict inequality.
      # And since the algorithm is to take the largest position with
      # maximum score, there is a possibility that we'll break on a found
      # score before but for splitting on a score after.
      LFBFL_j=$((LFBFL_i + 1))
      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_j} 1
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_j}

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      if update_position_score ${LFBFL_i}
      then
        break
      fi
    done
  else
    for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
      ((++LFBFL_i))
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
    done
  fi
```

```
  declare -ir LFBFL_i_max_score=$(
    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
  )
  local LFBFL_value
  declare -i LFBFL_i_best_position
  for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
    if [[ LFBFL_i -eq 0 ]]; then
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
      LFBFL_i_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=new_lines
  # $2=prefix
  # $3=max_length
  # $4=suffix
  # $5=split_score_command
  # $6=split_score_command_properties
  # $7=forbidden_previous_character ('\' usually)
  # Options:
  #   --verbose
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
```

```
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_length2=$(($3 - ${#4}))
  get_overlength_regexp "$3"

  # Testing if some line is too long.
  printf "%s" "$1"\
    | grep --quiet "${get_overlength_regexp_result}"
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
```

```
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.31 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  # $1=length
  # Option:
  #   --length-is-overlength
  declare -i LFBFL_i_overlength=$1
  declare -i LFBFL_i_length_is_overlength
  get_some_flag LFBFL_i_length_is_overlength --length-is-overlength 1 "$@"
  if [[ LFBFL_i_length_is_overlength -eq 0 ]]; then
    ((++LFBFL_i_overlength))
  fi
```

```
  readonly LFBFL_i_overlength
  declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result="^.\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"

  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_max_line_length}

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_path
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "too_long_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
```

```
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep --line-number --with-filename\
        --regexp="${get_overlength_regexp_result}" --
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.32 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

cd .. || exit
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./build_and_checks_dependencies/overwrite_if_not_equal.libr.sh"
cd build_and_checks_dependencies || exit

update_common_build_and_checks(){
  # Option:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
```

```
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp -- "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_dependencies_notes.exec.php"
    "build_dependencies_notes.libr.php"
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
```

```
      | cut --delimiter=' ' --fields=1
    )
    LFBFL_file_sha512_all="LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
    sed --in-place\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut --delimiter=' ' --fields=1
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed --in-place\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.33 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
50    declare -a LFBFL_return_traps_stack
51    init_return_trap
52    pushd_to_work_directory --trap-popd
53    can_continue_after_enhanced_pushd || return 1
54
55    declare -i LFBFL_i_write
56    get_some_flag LFBFL_i_write --write 1 "$@"
57    declare -i LFBFL_i_append
58    get_some_flag LFBFL_i_append --append 1 "$@"
59
60    enhanced_set_shell_option pipefail --trap-unset
61
62    LFBFL_subdir2="build_and_checks_variables"
63    declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
64    declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"
65
66    declare -i LFBFL_i_max_line_length
67    grep_variable "${LFBFL_data_file_path}" max_line_length\
68      --result-variable-prefix="LFBFL_i_"
69
70    if [[ LFBFL_i_write -eq 1 ]]; then
71      : > "${LFBFL_listing}"
72    fi
73    # Remove line returns that are here to keep lines short.
74    sed --regexp-extended --null-data --expression='s/(.)\\\n/\1/Mg'\
75      -- "${LFBFL_listing}"\
76      > "${LFBFL_listing}.temp"
77    # shopt -s dotglob was needed at some point but I don't see why now.
78    # shellcheck disable=SC2248
79    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
80    declare -r LFBFL_split_score_command="${get_split_score_result}"
81    declare -i LFBFL_i_split_score_command_properties
82    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
83    readonly LFBFL_i_split_score_command_properties
84    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
85    readonly LFBFL_suffix
86    # There is always a $'\n' final suffix from >>.
87    # But when origin string ends with the suffix we must add something.
88    declare -r LFBFL_final_suffix=$'\n\\'
89
90    local LFBFL_s_files_paths
91    LFBFL_s_files_paths=$(
92      find . -type f -printf '%P\n'\
93      | relevant_find\
94      | sort
95    )
```

```
  # printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf\
        "update_or_check_files_names_listing: Some files to process.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore --quiet -- "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
```

```
    fi
    if ! grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi

  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    if [[ ! -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm -- "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat -- "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

3.34 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts
```

3.35 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

3.36 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

3.37 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

3.38 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
  ] },
  { "dependency_name": "GNU coreutils", "URLs": [
      "https://www.gnu.org/software/coreutils/",
      "https://savannah.gnu.org/projects/coreutils/"
  ] },
  { "dependency_name": "uutils coreutils", "URLs": [
      "https://uutils.github.io/coreutils/",
      "https://github.com/uutils/coreutils"
  ] },
  { "dependency_name": "diff (GNU diffutils)", "URLs": [
      "https://www.gnu.org/software/diffutils/",
      "https://savannah.gnu.org/projects/diffutils/"
  ] },
  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
      "https://www.gnu.org/software/findutils/",
      "https://savannah.gnu.org/projects/findutils/"
  ] },
  { "dependency_name": "grep (GNU grep)", "URLs": [
      "https://www.gnu.org/software/grep/",
      "https://savannah.gnu.org/projects/grep/"
  ] },
  { "dependency_name": "sed (GNU sed)", "URLs": [
      "https://www.gnu.org/software/sed/",
      "https://savannah.gnu.org/projects/sed/"
  ] },
```

```
31    { "dependency_name": "ShellCheck", "URLs": [
32        "https://www.shellcheck.net/",
33        "https://github.com/koalaman/shellcheck"
34    ] },
35    { "dependency_name": "GNU Wget", "URLs": [
36        "https://www.gnu.org/software/wget/",
37        "https://savannah.gnu.org/projects/wget/"
38    ] },
39    { "dependency_name": "more of util-linux", "URLs": [
40        "https://www.kernel.org/pub/linux/utils/util-linux/",
41        "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42        {
43          "URL": "https://github.com/util-linux/util-linux",
44          "comment": "official?"
45        }
46    ] },
47    { "dependency_name": "pandoc", "URLs": [
48        "https://pandoc.org/",
49        "https://github.com/jgm/pandoc"
50    ] },
51    { "dependency_name": "pdftex/pdflatex", "URLs": [
52        "https://www.tug.org/applications/pdftex/",
53        "https://tug.org/texlive/svn/"
54    ] },
55    { "dependency_name": "tree", "URLs": [
56        "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57        "https://gitlab.com/OldManProgrammer/unix-tree",
58        "https://github.com/Old-Man-Programmer/tree"
59    ] },
60    { "dependency_name": "git", "URLs": [
61        "https://git-scm.com/about",
62        "https://github.com/git/git"
63    ] },
64    { "dependency_name": "pylint", "URLs": [
65        "https://www.pylint.org/",
66        "https://pylint.readthedocs.io/en/stable/",
67        "https://github.com/pylint-dev/pylint",
68        "https://pypi.org/project/pylint/"
69    ] },
70    { "dependency_name": "notify-send", "URLs": [
71        "https://gitlab.gnome.org/GNOME/libnotify/",
72        "https://gnome.pages.gitlab.gnome.org/libnotify/"
73    ] },
74    { "dependency_name": "black", "URLs": [
75        "https://github.com/psf/black",
76        "https://black.readthedocs.io/en/stable/",
```

```
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91          "https://phpdoc.org/",
92          "https://docs.phpdoc.org/",
93          "https://github.com/phpdocumentor/phpdocumentor"
94      ] },
95      { "dependency_name": "pip", "URLs": [
96          "https://pip.pypa.io/en/stable/",
97          "https://github.com/pypa/pip",
98          {
99            "URL": "https://pypi.org/project/pip/",
100           "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "xmllint (libxml2)", "URLs": [
109         "https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home",
110         "https://gitlab.gnome.org/GNOME/libxml2"
111     ] },
112     { "dependency_name": "bandit", "URLs": [
113         "https://bandit.readthedocs.io/en/latest/",
114         "https://github.com/PyCQA/bandit",
115         "https://pypi.org/project/bandit/"
116     ] },
117     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
118         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
119         "https://packagist.org/packages/squizlabs/php_codesniffer"
120     ] },
121     { "dependency_name": "phpmd", "URLs": [
122         "https://phpmd.org/",
```

```
        "https://github.com/phpmd/phpmd",
        "https://packagist.org/packages/phpmd/phpmd"
    ] },
    { "dependency_name": "php", "URLs": [
        "https://www.php.net/",
        "https://github.com/php/php-src"
    ] },
    { "dependency_name": "tex", "URLs": [
        "https://ctan.org/tex-archive/systems/knuth/dist/tex"
    ] },
    { "dependency_name": "pcregrep, pcre2grep", "URLs": [
        "https://www.pcre.org/",
        "https://pcre2project.github.io/pcre2/",
        "https://github.com/PCRE2Project/pcre1",
        "https://github.com/PCRE2Project/pcre2",
        "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
    ] },
    { "dependency_name": "python3", "URLs": [
        "https://www.python.org/",
        "https://docs.python.org/3/"
    ] }
]
```

3.39 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

```

```
grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
```

```
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/
https://docs.phpdoc.org/
https://github.com/phpdocumentor/phpdocumentor

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home
https://gitlab.gnome.org/GNOME/libxml2

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

```

```
phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.40 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.
```

```
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 -->
27 <head>
28 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
29 <meta name="generator" content="DevOrSysAdminScripts">
30 <style>
31 .title{
32   margin: 2ex auto;
33   text-align: center;
34 }
35 .titlemain{
36   margin: 1ex 2ex 2ex 1ex;
37 }
38 .titlerest{
39   font-size: 1.17em;
40   margin: 0ex 2ex;
41 }
42 table{
43   border-collapse: collapse;
44 }
45 td{
46   padding: 0;
47 }
48 pre{
49   text-align: left;
50   margin-left: 0ex;
51   margin-right: auto;
52 }
53 pre.numbered_lines:before{
54   background: snow;
55   border-right: 1px black solid;
56   color: #010101;
57   content: attr(line-numbers);
58   float: left;
59   padding: 0 1ch;
60   text-align: right;
61 }
62 blockquote{
63   margin-left: 4ex;
64   margin-right: 4ex;
```

```
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
```

```
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files trees</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files trees</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
```

```
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, phpdoc, pip, ruff, xmllint;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>
```

```
<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.41 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
```

```
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
```

```
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files trees}
\label{section:trees}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}
```

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, phpdoc, pip, ruff, xmllint;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

```

```
\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.42 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
```

```
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
```

```
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234 5
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890 123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
```

```
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.43 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
```

```
whilst GNU coreutils is GPLv3+ licensed.
```

3.44 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
  I added all the standards, then excluded all the rules that gave me
  false positives or that seemed unappropriate when looking at the
  documentation.
  Some are bugs, some are conflicting with my own coding style.
  You may delete an exclude and see what pops up to maybe understand
  my choices.
  -->

  <rule ref="Generic">
    <!-- You have to allow long or short or both syntaxes. -->
    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
    <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
    <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
    <!--
    Not standard in Unix/Linux world to keep files not ending with \n.
    This is always an option of the code editors.
    Since PHP doesn't output the final \n after ?>, it is ok.
    -->
    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
    <exclude name="Generic.Files.ExecutableFile.Executable"/>
    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
    <exclude
     name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
    />
    <exclude
      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
```

```
43      />
44      <!-- Applies only to true, false and null. -->
45      <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
46      <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
47      <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
48      <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
49    </rule>
50    <rule ref="PEAR">
51      <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
52      <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
53      <!--
54      Look at the comment before generate_split_score.
55      PEAR and Squiz require to align the parameter names parts in the same
56      comment, and also the parameter comments parts in the same comment,
57      but with a long parameter name like $a_delimiter_strings_domain
58      and a line length limit at 75 characters, parameter comments parts
59      have only 24 characters left per line, which is ridiculous.
60      Our choice to put the parameter comments parts starting on another line
61      with same "indent" as parameter names parts is way better in those
62      cases. Alignment has always been a trap with short lines.
63      Since phpDocumentor handles the parameter comments parts correctly,
64      it is enough for me.
65      Worse is the handling of "nested doc block comment" for the Closure
66      returned by generate_split_score().
67      Our solution makes at least the source code readable.
68      Maybe one day the dev tools will handle it correctly.
69      -->
70      <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
71      <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
72      <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
73      <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
74      <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
75      <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
76      <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
77      <exclude
78        name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
79      />
80      <exclude
81       name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
82      />
83      <exclude
84        name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
85      />
86      <!--
87      Don’t laugh when looking at the false positives of the two exclude
88      below ;) XD.
```

```
    They don't check for multiline function declarations apparently.
    -->
    <exclude
      name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
    />
    <exclude
      name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
    />
    <exclude
      name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
    />
    <exclude
      name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
    />
  </rule>
  <rule ref="PSR1">
  </rule>
  <rule ref="PSR2">
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
    <exclude
name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
    />
  </rule>
  <rule ref="PSR12">
    <exclude
 name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
    />
    <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
    <!--
    3 false positives, see:
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
    -->
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
    />
    <!-- and even a fourth and fifth false positives -->
    <exclude
      name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
    />
```

```
    <exclude
      name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
    />
    <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
    <exclude
      name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
    />
    <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
    <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
    <!-- I agree except for string concatenation. -->
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
  </rule>
  <rule ref="Squiz">
    <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
    <!--
    It doesn't know the //*/ trick...
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
    -->
    <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
    <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
    <!-- Problem for doc block comment before namespace -->
    <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
    <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
    <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
    <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
    <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
    <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
    <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude
      name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
    />
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
    <exclude
```

```
      name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
    />
    <exclude
      name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
    />
    <!-- Problem for doc block comment before namespace -->
    <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
    <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
    <!-- It doesn't know the //*/ trick... -->
    <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
    <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
    <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
    <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
    <exclude
      name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
    />
    <exclude
  name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
    />
    <exclude
      name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
    />
    <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
    <exclude name="Squiz.Functions.GlobalFunction.Found"/>
    <exclude
      name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
    />
    <exclude
  name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
    />
    <exclude
      name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
    />
    <exclude
      name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
    />
    <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
    <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
    <exclude name="Squiz.PHP.CommentedOutCode"/>
    <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
    <exclude
      name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
    />
```

```
227     </rule>
228     <rule ref="Zend">
229       <!-- Back to my habits at Teliae with always a closing tag. -->
230       <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
231       <exclude
232         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
233       />
234     </rule>
235
236     <rule ref="Generic.Arrays.ArrayIndent">
237       <properties>
238         <property name="indent" value="2" />
239       </properties>
240     </rule>
241     <rule ref="Generic.Files.LineLength">
242       <properties>
243         <property name="lineLimit" value="75"/>
244         <property name="absoluteLineLimit" value="75"/>
245       </properties>
246     </rule>
247     <rule ref="Generic.Formatting.SpaceAfterNot">
248       <properties>
249         <property name="spacing" value="0" />
250       </properties>
251     </rule>
252     <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
253       <properties>
254         <property name="checkFunctions" value="true"/>
255         <property name="checkClosures" value="true"/>
256       </properties>
257     </rule>
258     <rule ref="Generic.Metrics.CyclomaticComplexity">
259       <properties>
260         <property name="complexity" value="40" />
261         <property name="absoluteComplexity" value="80" />
262       </properties>
263     </rule>
264     <rule ref="Generic.Strings.UnnecessaryStringConcat">
265       <properties>
266         <property name="allowMultiline" value="true" />
267       </properties>
268     </rule>
269     <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
270       <properties>
271         <property name="ignoreNewlines" value="true" />
272       </properties>
```

```
273     </rule>
274     <rule ref="Generic.WhiteSpace.ScopeIndent">
275       <properties>
276         <property name="ignoreIndentationTokens" type="array">
277           <element value="T_COMMENT"/>
278           <element value="T_DOC_COMMENT_OPEN_TAG"/>
279         </property>
280         <property name="indent" value="2"/>
281       </properties>
282     </rule>
283
284     <rule ref="PEAR.Functions.FunctionCallSignature">
285       <properties>
286         <property name="indent" value="2"/>
287       </properties>
288     </rule>
289     <rule ref="PEAR.Functions.FunctionDeclaration">
290       <properties>
291         <property name="indent" value="2"/>
292       </properties>
293     </rule>
294     <rule ref="PEAR.WhiteSpace.ScopeIndent">
295       <properties>
296         <property name="ignoreIndentationTokens" type="array">
297           <element value="T_COMMENT"/>
298           <element value="T_DOC_COMMENT_OPEN_TAG"/>
299         </property>
300         <property name="indent" value="2"/>
301       </properties>
302     </rule>
303
304     <rule ref="PSR2.Methods.FunctionCallSignature">
305       <properties>
306         <property name="indent" value="2"/>
307       </properties>
308     </rule>
309
310     <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
311       <properties>
312         <property name="indent" value="2"/>
313       </properties>
314     </rule>
315
316     <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
317       <properties>
318         <property name="indent" value="2"/>
```

```
    </properties>
  </rule>
  <rule ref="Squiz.Strings.ConcatenationSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Squiz.WhiteSpace.FunctionSpacing">
    <properties>
      <property name="spacing" value="3" />
      <property name="spacingBeforeFirst" value="0" />
      <property name="spacingAfterLast" value="0" />
    </properties>
  </rule>
</ruleset>
```

3.45 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"
```

```
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"
  />
  <!-- https://github.com/phpmd/phpmd/issues/910 -->
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"
  />
</phpmd-baseline>
```

3.46 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
```

```
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.47 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
```

```
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
```

```
          }
        },
        "D": {
          "0.1.0": {
            "C": {
              // Dumb, you did not found it by yourself? Failed exam.
              "4.4.4" : {}
            }
          }
        }
      }
    },
    "B": {
      "2.2.2": {
        "C": {
          // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
  }
  Wonderful someone rediscovered nested structures XD
  Seriously so many engineers or computer scientists are dumb to cry.

  bandit_venv=../venv_bandit
  black_venv=../venv_black
  isort_venv=../venv_isort
  mypy_venv=../venv_mypy
  pylint_venv=../venv_pylint
  ruff_venv=../venv_ruff


  upgrade venvs can be auto or ask, any other value is interpreted
  as ask.

  upgrade_venvs=ask


  time interval can be 0 to always upgrade or ask to upgrade,
  or a number at most bash max int
  https://superuser.com/questions/1030122/
  what-is-the-maximum-value-of-a-numeric-bash-shell-variable
  it should be enough to simulate never "upgrade or ask for upgrade"
  unless you claim to live many billions years ;)
  If it is wat, this is a random int XD (see RANDOM).
  Note that the probability that it upgrades your dependencies is not
```

```
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false

xmllint_files_without_DTD=./build_and_checks_variables/phpcs_ruleset.xml
xmllint_files_without_DTD=./build_and_checks_variables/phpmd_baseline.xml
```

3.48 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

```

```
acknowledgments=
acknowledgments=

max_line_length=

latex_command=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=

xmllint_files_without_DTD=
xmllint_files_without_DTD=
```

3.49 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD
```

```

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.50 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.51 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
```

```
            Use a directive to specify location
            # shellcheck source=bla/bli.sh
            Only remaining cases are in common_build_and_checks
            when sourcing some Python venv
            source venv/bin/active
- SC1091 : Not following (file not found most of the time).
            Only remaining cases are in common_build_and_checks
            when sourcing some Python venv
            source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
            False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
            False positive when c contains \n.
- SC2016 : Expressions don't expand in single quotes,
            use double quotes for that.
- SC2034 : Unused variable
            False positive for globals that are declared in functions
            that are not currently called.
            Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
            Use printf '..%s..' "$foo".
            False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
            False positive when returning integer variable
            via an indirection like return ${!foo}.
            See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
            False positive when passing arguments would yield
            a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
            False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
            Very useful but false positive when I play with
            variable variable names.
- SC2162X: read without -r will mangle backslashes.
            False positive when read is used without variable:
```

```
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
              Use = to compare as string (or expand explicitly with $var).
            Very useful but false positive when I play with
            variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
            https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$"\
  -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$"\
  -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.52 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
```

```
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
  mkdir --parents -- "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

```

```
71    local LFBFL_dependencies_URL
72    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74    readonly LFBFL_dependencies_URL
75
76    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77    local LFBFL_script_download_URL
78    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79    LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80    readonly LFBFL_script_download_URL
81    declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82    local LFBFL_correct_sha512
83    LFBFL_correct_sha512='9e92750cda51faf49210124fec7fe8a2554013ebb1295'
84    LFBFL_correct_sha512+='0d2e9c63897613ba3f233e953cd54c3bcd02dcabae19'
85    LFBFL_correct_sha512+='8ce1f933713dc900308cdfa23b7b3e8a44b00c2'
86    wget_sha512 "${LFBFL_file_path}"\
87      "${LFBFL_script_download_URL}"\
88      "${LFBFL_correct_sha512}"\
89      "${LFBFL_verbose}"\
90      "${LFBFL_check_download}"
91    chmod +x "${LFBFL_file_path}"
92
93    "${LFBFL_file_path}" "${LFBFL_work_directory}"\
94      "${LFBFL_dependencies_URL}"\
95      "${LFBFL_verbose}"\
96      "${LFBFL_check_download}"
97  }
98
99  if [[ "$*" == *--fixed-point-build* ]]; then
100   printf -- "--fixed-point-build\n"
101   source "./build_and_checks_dependencies/lines_filters.libr.sh"
102   LFBFL_work_directory="."
103   if [[ -n "$1" ]]; then
104     LFBFL_work_directory="$1"
105   fi
106   LFBFL_variables_directory="${LFBFL_work_directory}/"
107   LFBFL_variables_directory+="build_and_checks_variables"
108   LFBFL_data_file_name="${LFBFL_variables_directory}/"
109   LFBFL_data_file_name+="repository_data.txt"
110   LFBFL_repository_name=""
111   grep_variable "${LFBFL_data_file_name}" repository_name\
112     --result-variable-prefix="LFBFL_"\
113     --replace-line-returns-by=""
114   # Touching the 3 following files first let us iterate
115   # upgrade_build_and_checks 2 times instead of 3 to reach
116   # a fixed point.
```

```
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.53 COPYING

```
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```
    copy of the Corresponding Source for all the software in the
    product that is covered by this License, on a durable physical
    medium customarily used for software interchange, for a price no
    more than your reasonable cost of physically performing this
    conveying of source, or (2) access to copy the
    Corresponding Source from a network server at no charge.

    c) Convey individual copies of the object code with a copy of the
    written offer to provide the Corresponding Source.  This
    alternative is allowed only occasionally and noncommercially, and
    only if you received the object code with such an offer, in accord
    with subsection 6b.

    d) Convey the object code by offering access from a designated
    place (gratis or for a charge), and offer equivalent access to the
    Corresponding Source in the same way through the same place at no
    further charge.  You need not require recipients to copy the
    Corresponding Source along with the object code.  If the place to
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```

```
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```

```
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```

```
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```

3.55 git_helpers/git_checks.libr.sh

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
```

```
61    fi
62    if grep serializer /tmp/DOSAS_django_git_check1.temp;
63    then
64      notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65      LFBFL_i_error=1
66    fi
67    rm /tmp/DOSAS_django_git_check1.temp
68    # shellcheck disable=SC2248
69    return ${LFBFL_i_error}
70  }
71
72
73  check_no_abusive_trailing_comma(){
74    # Useful for Python code, although it regularly yields
75    # false positives in Python code
76    # (but it is easy to distinguish by hand).
77    # Use `git commit --no-verify` after check.
78    declare -r LFBFL_send_summary_1="ATTENTION"
79    local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80    LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81    readonly LFBFL_send_body_1
82    declare -i LFBFL_i_error=0
83    git diff --cached\
84      > /tmp/DOSAS_django_git_check2.temp
85    if grep --perl-regexp ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86    then
87      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88      LFBFL_i_error=1
89    fi
90    rm /tmp/DOSAS_django_git_check2.temp
91    # shellcheck disable=SC2248
92    return ${LFBFL_i_error}
93  }
94
95
96  check_black_code_formatting(){
97    # attention ça ne marche que sur les fichiers "stagés" avec git add
98    declare -r LFBFL_s_files_paths=$(
99      git diff --cached --name-only\
100     | grep '\.py'
101   )
102   printf "%s\n" "${LFBFL_s_files_paths}"
103   declare -a LFBFL_arr_files_paths
104   mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
105   readonly LFBFL_arr_files_paths
106   declare -i LFBFL_i_error=0
```

```
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
    if ! black --check --diff -- "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.56 git_helpers/post-commit

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

```

```
LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.57 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# or (at your option) any later version.
```

```
#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd -- "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf\
    "post_commit_renaming_comment.libr.sh pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
```

```
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -i LFBFL_i_log_directory_change
  get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
  declare -i LFBFL_i_max_comment_line_length
  get_some_option LFBFL_i_max_comment_line_length\
    --max-comment-line-length 70 70 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  if [[ LFBFL_i_max_comment_line_length -lt 10 ]]; then
    declare -r LFBFL_send_summary="Max comment line length below 10."
    notify-send "${LFBFL_send_summary}" "${LFBFL_send_summary}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "Max comment line length below 10, aborting.\n"
    fi
    return
  fi
  local LFBFL_s_format
  #-----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

```

```
102    declare -r LFBFL_renaming_lines=$(
103      git log --max-count=1 --patch\
104      | grep --after-context=1 '^diff --git'
105    )
106    if [[ LFBFL_i_verbose -eq 1 ]]; then
107      printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
108    fi
109
110    declare -a LFBFL_renaming_lines_array
111    mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
112    readonly LFBFL_renaming_lines_array
113
114    declare -r LFBFL_timestamp=$(date --iso-8601="minute")
115    if [[ LFBFL_i_verbose -eq 1 ]]; then
116      printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
117    fi
118    local LFBFL_split_score_command1
119    declare -i LFBFL_i_split_score_command_properties1
120    # shellcheck disable=SC2248
121    get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
122    LFBFL_split_score_command1="${get_split_score_result}"
123    LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
124    readonly LFBFL_split_score_command1
125    readonly LFBFL_i_split_score_command_properties1
126    local LFBFL_split_score_command2
127    declare -i LFBFL_i_split_score_command_properties2
128    # shellcheck disable=SC2248
129    get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
130    LFBFL_split_score_command2="${get_split_score_result}"
131    LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
132    readonly LFBFL_split_score_command2
133    readonly LFBFL_i_split_score_command_properties2
134    declare -i LFBFL_i_renaming_happened=0
135    declare -i LFBFL_i
136    declare -i LFBFL_j
137    declare -r LFBFL_special_file_name="files_names_listing.txt"
138    local LFBFL_diff_line
139    local LFBFL_similarity_line
140    local LFBFL_old_file_path
141    local LFBFL_old_file_name
142    local LFBFL_old_file_directory
143    local LFBFL_new_file_path
144    local LFBFL_new_file_name
145    local LFBFL_new_file_directory
146    local LFBFL_useful_file_name
147    local LFBFL_extension
```

```
148     local LFBFL_comment_prefix
149     local LFBFL_comment_prefix2
150     local LFBFL_old_file_name2
151     local LFBFL_old_file_directory2
152     local LFBFL_new_file_name2
153     local LFBFL_new_file_directory2
154     local LFBFL_new_comment
155     declare -i LFBFL_i_copyright_line_number
156     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
157     declare -i LFBFL_i_line_count
158     declare -i LFBFL_i_lines_after
159     local LFBFL_temp_file_path
160     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
161       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
162       LFBFL_i=$((LFBFL_i + 1))
163       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
164       LFBFL_i=$((LFBFL_i + 1))
165       # We need to increment 3 times since grep adds "--" lines between.
166       LFBFL_i=$((LFBFL_i + 1))
167       if [[ LFBFL_i_verbose -eq 1 ]]; then
168         printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
169           "${LFBFL_diff_line}"\
170           "${LFBFL_similarity_line}"
171       fi
172       if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
173         continue
174       fi
175
176       # Extract old file name from diff line. --------------------------
177       LFBFL_old_file_path=$(
178         printf "%s" "${LFBFL_diff_line}"\
179         | cut --delimiter=' ' --fields=3
180       )
181       # It starts by "a/".
182       LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
183       LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
184       LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
185       # Extract new file name from diff line. --------------------------
186       LFBFL_new_file_path=$(
187         printf "%s" "${LFBFL_diff_line}"\
188         | cut --delimiter=' ' --fields=4
189       )
190       # It starts by "b/".
191       LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
192       LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
193       LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")
```

```
194     if [[ LFBFL_i_verbose -eq 1 ]]; then
195       LFBFL_s_format="LFBFL_old_file_path: %s\n"
196       LFBFL_s_format+="LFBFL_old_file_name: %s\n"
197       LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
198       LFBFL_s_format+="LFBFL_new_file_path: %s\n"
199       LFBFL_s_format+="LFBFL_new_file_name: %s\n"
200       LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
201       # shellcheck disable=SC2059
202       printf "${LFBFL_s_format}"\
203         "${LFBFL_old_file_path}"\
204         "${LFBFL_old_file_name}"\
205         "${LFBFL_old_file_directory}"\
206         "${LFBFL_new_file_path}"\
207         "${LFBFL_new_file_name}"\
208         "${LFBFL_new_file_directory}"
209     fi
210     # Extract "LFBFL_extension" with smart guessing for .tpl ---------
211     # and set prefix for line with ©Copyright. -----------------------
212     LFBFL_useful_file_name="${LFBFL_new_file_name}"
213     LFBFL_extension="${LFBFL_useful_file_name##*.}"
214     if [[ LFBFL_i_verbose -eq 1 ]]; then
215       printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
216         "${LFBFL_useful_file_name}"\
217         "${LFBFL_extension}"
218     fi
219     for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
220       if [[ "${LFBFL_extension}" != "tpl" ]]; then
221         break
222       fi
223       LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
224       LFBFL_extension="${LFBFL_useful_file_name##*.}"
225       if [[ LFBFL_i_verbose -eq 1 ]]; then
226         printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
227           "${LFBFL_useful_file_name}"\
228           "${LFBFL_extension}"
229       fi
230     done
231     LFBFL_comment_prefix=""
232     if [[ "${LFBFL_extension}" == "sh" ]]; then
233       LFBFL_comment_prefix="# "
234     fi
235     if [[ "${LFBFL_extension}" == "sql" ]]; then
236       LFBFL_comment_prefix="-- "
237     fi
238     if [[ "${LFBFL_extension}" == "tex" ]]; then
239       LFBFL_comment_prefix="% "
```

```
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_special_file_name}" ]];
    then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
```

```
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[
      LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
    ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_i_copyright_line_number=$(
```

```
      grep --line-number -- '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut --delimiter=':' --fields=1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. -------------------------------------
    LFBFL_i_line_count=$(
      wc --lines\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. ---------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        -- "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add -- "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
```

```
    # Commit -------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.58 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
```

```
  LFBFL_i_pushd_result3=$?
  printf "pre-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.59 IMAP_transfert.py

```
"""
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```

```
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
```

```
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
```

```
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print('Connecting to "from account".')
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print('Creating new folder in "to account".')
        to_account.create(to_new_folder)

    if verbose:
        print('Fetching all emails ids in "from account".')
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print('Fetching and transfering each email in "from account".')
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
```

```
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
```

```
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
```

```
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

3.60 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
```

```
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
https://pubs.opengroup.org/onlinepubs/9699919799/
When looking `man sed` GNU sed says to use POSIX -E instead of
-r or --regexp-extended for portability.
But the link above says that still now only -e, -f, -n are POSIX options
for sed.
For this reason we replaced all short option names by long option names
when possible, even for sed.
```

3.61 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
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# or (at your option) any later version.
#
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# that it will be useful,
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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# along with DevOrSysAdminScripts.
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#
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
```

```
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.62 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.63 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
```

```
</pre>

<script>
let a = 'abcd & < > \\ \n  \"\'';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.64 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.65 README.md.tpl

```
# DevOrSysAdminScripts

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[![Security:bandit:badge]][Bandit-GitHub.com]
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[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
```

```
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

```

```
[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.66 string_escaping.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts library.

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```

```

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@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for HTML (surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\HTML
*/
namespace string_escaping\HTML {
  /**
  When you want that the string appears as is in HTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()


```

```
  /**
  When you want that the string appears as is in an HTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      ['"'],
      ['&quot;'],
      $s_string,
    );
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside HTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\HTML
```

```
/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for XHTML (still surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\XHTML
*/
namespace string_escaping\XHTML {
  /**
  When you want that the string appears as is in XHTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an XHTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      [
        '&',
```

```
        '<',
        '"',
        "\n",
      ],
      [
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;',
      ],
      $s_string,
    );
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside XHTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\XHTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for JS (you're hopeless?)...

```

```
@package DevOrSysAdminScripts
@subpackage string_escaping\JS
*/
namespace string_escaping\JS {

  use function string_escaping\XHTML\escape_text as XHTML_escape_text;



  /**
  When you want that the string can be put in a JS string definition.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape(string $s_string) : string {
    return str_replace(
      [
        '\\',
        "\n",
        "'",
        '"',
      ],
      [
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ],
      $s_string,
    );
  }//end escape()



  /**
  When you want that the string can be put in a JS string definition
  to be included inside an XHTML file <script> tag.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
```

```
    return XHTML_escape_text(escape($s_string));
  }//end escape_for_string_definition_inside_XHTML()
}//end namespace string_escaping\JS




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for TeX.

@package DevOrSysAdminScripts
@subpackage string_escaping\TeX
*/
namespace string_escaping\TeX {
  /**
  When you want that the string can be written to a TeX file inside a
  part where TeX is in text mode.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    // https://tex.stackexchange.com/a/34586/270015
    return str_replace(
      [
        '\\',
        '&',
        '%',
        '$',
        '#',
        '_',
        '{',
        '}',
        '~',
        '^',
      ],
      [
        '\\textbackslash',
        '\\&',
        '\\%',
        '\\$',
        '\\#',
        '\\_',
```

```
        '\\{',
        '\\}',
        '\\textasciitilde',
        '\\textasciicircum',
      ],
      $s_string,
    );
  }//end escape_text()
}//end namespace string_escaping\TeX

/*
$ php
<?php
require_once('string_escaping.libr.php');
var_dump(string_escaping\HTML\escape_text("<&\"\n'truc\\>"));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\>"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_text("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\>"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
?>
```

3.67 unicode.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts library.

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you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
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```

```
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This file was renamed from "unicode.php" to "unicode.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
decimal_code_point is a slight abuse of language to say that you write the
code point in decimal notation in PHP code and you get the UTF-8.
If you provide the integer of the code point without using decimal
notation, it will also work.

@param int $i_code_point_in_decimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
```

```
64    if($i_code_point_in_decimal_notation < 2048){  // 2**11
65      $i_first_byte_base_value = 192;
66      /*
67      $i_last_byte_significant_bits = (
68        $i_code_point_in_decimal_notation % 64  // 2**6
69      );
70      $i_first_byte_significant_bits = intdiv(
71        (
72          $i_code_point_in_decimal_notation
73          - $i_last_byte_significant_bits
74        ),
75        64,  // 2**6
76      );
77      $s_result = chr(
78        $i_first_byte_base_value
79        + $i_first_byte_significant_bits
80      );
81      $s_result .= chr(
82        $i_continuation_base_value
83        + $i_last_byte_significant_bits
84      );
85      return $s_result;
86      */
87      return (
88        chr(
89          $i_first_byte_base_value
90          + ($i_code_point_in_decimal_notation >> 6)
91        )
92        .chr(
93          $i_continuation_base_value
94          + ($i_code_point_in_decimal_notation & 63)
95        )
96      );
97    }//end if($i_code_point_in_decimal_notation < 2048)
98    // 1110xxxx 10xxxxxx 10xxxxxx
99    if($i_code_point_in_decimal_notation < 65536){  // 2**16
100     $i_first_byte_base_value = 224;
101     /*
102     $arr_arr_data_per_byte_reverse = [
103       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
104       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
105       ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
106     ];
107     $i_rest = $i_code_point_in_decimal_notation;
108     $s_result = '';
109     foreach($arr_arr_data_per_byte_reverse as $arr_data){
```

```
110         // _Pragma unroll
111         // J'aurais pu "roller" les 4 cas des ifs.
112         // Mais pour les perfs, il faudrait tout unroller vu le nombre
113         // d'itérations. Et j'aimerais bien comparer du code unrollé
114         // aux 2 niveaux à la main avec celui unrollé par le compilateur
115         // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
116         // l'intérieur d'abord (pour ne pas refaire le même boulot)
117         // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
118         // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
119         $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
120         $i_rest = intdiv(
121           $i_rest - $i_significant_bits,
122           2**$arr_data['i_bits'],
123         );
124         // .= or =.
125         // Below could have been a case of "=.".
126         $s_result = (
127           chr($i_significant_bits + $arr_data['i_base_value'])
128           .$s_result
129         );
130       }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
131       return $s_result;
132       */
133       return (
134         chr(
135           $i_first_byte_base_value
136           + ($i_code_point_in_decimal_notation >> 12)
137         )
138         .chr(
139           $i_continuation_base_value
140           + (($i_code_point_in_decimal_notation >> 6) & 63)
141         )
142         .chr(
143           $i_continuation_base_value
144           + ($i_code_point_in_decimal_notation & 63)
145         )
146       );
147     }//end if($i_code_point_in_decimal_notation < 65536)
148     // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
149     if($i_code_point_in_decimal_notation < 2097152){  // 2**21
150       $i_first_byte_base_value = 240;
151       /*
152       $arr_arr_data_per_byte_reverse = [
153         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
154         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
155         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
```

```
      ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }
    return $s_result;
    */
    return (
      chr(
        $i_first_byte_base_value
        + ($i_code_point_in_decimal_notation >> 18)
      )
      .chr(
        $i_continuation_base_value
        + (($i_code_point_in_decimal_notation >> 12) & 63)
      )
      .chr(
        $i_continuation_base_value
        + (($i_code_point_in_decimal_notation >> 6) & 63)
      )
      .chr(
        $i_continuation_base_value
        + ($i_code_point_in_decimal_notation & 63)
      )
    );
  }//end if($i_code_point_in_decimal_notation < 2097152)
  // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
```

```
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
    ." :)"
  );
  // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
}//end decimal_code_point_to_UTF8()



/**
Converts a string of hexadecimal digits to the byte-string corresponding
to the UTF-8 encoding of the Unicode code point of same integer value
that is denoted by the hexadecimal digits.

@param string $s_code_point_in_hexadecimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function hexa_code_point_to_UTF8(
  string $s_code_point_in_hexadecimal_notation
) : string {
  $s_code_point_in_hexadecimal_notation = strtoupper(
    $s_code_point_in_hexadecimal_notation
  );
  $i_code_point_in_decimal_notation = 0;
  $i_ord_0 = ord('0');
  $i_ord_A = ord('A');
  $i_digit_offset = -$i_ord_0;
  $i_letter_offset = 10 - $i_ord_A;
  for(
    $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
    $i < $i_max;
    ++$i
  ){
    $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
    $i_code_point_in_decimal_notation *= 16;
    if(ctype_digit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_digit_offset
      );
      continue;
    }
    if(ctype_xdigit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
```

```
248         ord($s_hexa_digit) + $i_letter_offset
249       );
250       continue;
251     }
252     throw new Exception('Not an hexadecimal digit.');
253   }
254   return decimal_code_point_to_UTF8(
255     $i_code_point_in_decimal_notation
256   );
257 }//end hexa_code_point_to_UTF8()
258
259
260
261 /**
262 This function throws an exception with extended debug informations
263 if the input string is not valid ASCII.
264 Otherwise, it returns true.
265
266 @param string $s_string The input string.
267
268 @throws Exception When the input string is not valid ASCII.
269
270 @return bool
271 */
272 function check_string_is_valid_ASCII(string $s_string) : bool {
273   $i_offset_in_octets_from_string_start = 0;
274   $i_current_line_number = 1;
275   $i_offset_in_octets_from_line_start = -1;
276   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
277     $i_current_octet = ord($s_string[$i]);
278     $i_offset_in_octets_from_string_start = $i;
279     ++$i_offset_in_octets_from_line_start;
280     if($i_current_octet > 127){
281       throw new Exception(
282         'Non-ASCII character found on line '
283         .$i_current_line_number
284         .'; the octet/character '
285         .($i_offset_in_octets_from_line_start + 1)
286         .' has value '
287         .$i_current_octet
288         .' which is not ASCII.'
289         .' (Sequential position without line splitting:'
290         .' This is at octet/character '
291         .($i_offset_in_octets_from_string_start + 1)
292         .'.)',
293       );
```

```
294       }
295       // if($s_string[$i] === "\n"){
296       if($i_current_octet === 10){
297         ++$i_current_line_number;
298         $i_offset_in_octets_from_line_start = -1;
299       }
300     }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
301     return true;
302   }//end check_string_is_valid_ASCII()
303
304
305
306   /**
307   This function throws an exception with extended debug informations
308   if the content of the file with given file_path is not valid ASCII.
309   Otherwise, it returns true.
310
311   @param string $s_file_path The input file_path.
312
313   @throws Exception When the input string is not valid ASCII.
314
315   @return bool
316   */
317   function check_file_is_valid_ASCII(string $s_file_path) : bool {
318     $s_string = file_get_contents($s_file_path);
319     if($s_string === false){
320       throw new Exception('File '.$s_file_path.' not found.');
321     }
322     return check_string_is_valid_ASCII($s_string);
323   }//end check_file_is_valid_ASCII()
324
325
326
327   /**
328   This function throws an exception with extended debug informations
329   if the input string is not valid UTF-8.
330   Otherwise, it returns true.
331
332   @param string $s_string The input string.
333
334   @throws Exception When the input string is not valid UTF-8.
335
336   @return bool
337   */
338   function check_string_is_valid_UTF8(string $s_string) : bool {
339     $i_continuation_octet_needed = 0;
```

```
340     $i_offset_in_octets_from_string_start = 0;
341     $i_offset_in_characters_from_string_start = -1;
342     $i_character_start_position_from_string_start = 0;
343     $i_current_line_number = 1;
344     $i_offset_in_octets_from_line_start = -1;
345     $i_offset_in_characters_from_line_start = -1;
346     $i_character_start_position_from_line_start = -1;
347     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
348       $i_current_octet = ord($s_string[$i]);
349       $i_offset_in_octets_from_string_start = $i;
350       ++$i_offset_in_octets_from_line_start;
351       if($i_continuation_octet_needed > 0){
352         if($i_current_octet < 128 || $i_current_octet >= 192){
353           throw new Exception(
354             'Non-UTF8 character found on line '
355             .$i_current_line_number
356             .'; the octet '
357             .($i_offset_in_octets_from_line_start + 1)
358             .', part of the character '
359             .($i_offset_in_characters_from_line_start + 1)
360             .', has value '
361             .$i_current_octet
362             .' which is not a continuation octet.'
363             .' This character starts at octet '
364             .($i_character_start_position_from_line_start + 1)
365             .' of the current line.'
366             .' (Sequential positions without line splitting:'
367             .' This is at character '
368             .($i_offset_in_characters_from_string_start + 1)
369             .' and octet '
370             .($i_offset_in_octets_from_string_start + 1)
371             .'.'
372             .' This character starts at octet '
373             .($i_character_start_position_from_string_start + 1)
374             .'.)',
375           );
376         }//end if($i_current_octet < 128 || $i_current_octet >= 192)
377         --$i_continuation_octet_needed;
378         continue;
379       }//end if($i_continuation_octet_needed > 0)
380
381       ++$i_offset_in_characters_from_string_start;
382       ++$i_offset_in_characters_from_line_start;
383       $i_character_start_position_from_string_start = (
384         $i_offset_in_octets_from_string_start
385       );
```

```
386       $i_character_start_position_from_line_start = (
387         $i_offset_in_octets_from_line_start
388       );
389       if($i_current_octet < 128){ // 0xxxxxxx ASCII
390         // if($s_string[$i] === "\n"){
391         if($i_current_octet === 10){
392           ++$i_current_line_number;
393           $i_offset_in_octets_from_line_start = -1;
394           $i_offset_in_characters_from_line_start = -1;
395         }
396         continue;
397       }
398       if($i_current_octet >= 128 && $i_current_octet < 192){
399         throw new Exception(
400           'Non-UTF8 character found on line '
401           .$i_current_line_number
402           .'; the octet '
403           .($i_offset_in_octets_from_line_start + 1)
404           .', part of the character '
405           .($i_offset_in_characters_from_line_start + 1)
406           .', has value '
407           .$i_current_octet
408           .' which is a continuation octet.'
409           .' This character starts at octet '
410           .($i_character_start_position_from_line_start + 1)
411           .' of the current line.'
412           .' (Sequential positions without line splitting:'
413           .' This is at character '
414           .($i_offset_in_characters_from_string_start + 1)
415           .' and octet '
416           .($i_offset_in_octets_from_string_start + 1)
417           .'.'
418           .' This character starts at octet '
419           .($i_character_start_position_from_string_start + 1)
420           .'.)',
421         );
422       }//end if($i_current_octet >= 128 && $i_current_octet < 192)
423       if($i_current_octet >= 192 && $i_current_octet < 224){
424         // 110xxxxx 10xxxxxx
425         $i_continuation_octet_needed = 1;
426         continue;
427       }
428       if($i_current_octet >= 224 && $i_current_octet < 240){
429         // 1110xxxx 10xxxxxx 10xxxxxx
430         $i_continuation_octet_needed = 2;
431         continue;
```

```
432      }
433      if($i_current_octet >= 240 && $i_current_octet < 248){
434        // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
435        $i_continuation_octet_needed = 3;
436        continue;
437      }
438      throw new Exception(
439        'Non-UTF8 character found on line '
440        .$i_current_line_number
441        .'; the octet '
442        .($i_offset_in_octets_from_line_start + 1)
443        .', part of the character '
444        .($i_offset_in_characters_from_line_start + 1)
445        .', has value '
446        .$i_current_octet
447        .' which is invalid.'
448        .' This character starts at octet '
449        .($i_character_start_position_from_line_start + 1)
450        .' of the current line.'
451        .' (Sequential positions without line splitting:'
452        .' This is at character '
453        .($i_offset_in_characters_from_string_start + 1)
454        .' and octet '
455        .($i_offset_in_octets_from_string_start + 1)
456        .'.'
457        .' This character starts at octet '
458        .($i_character_start_position_from_string_start + 1)
459        .'.)',
460      );
461    }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
462    return true;
463  }//end check_string_is_valid_UTF8()
464
465
466
467  /**
468  This function throws an exception with extended debug informations
469  if the content of the file with given file_path is not valid ASCII.
470  Otherwise, it returns true.
471
472  @param string $s_file_path The input file_path.
473
474  @throws Exception When the input string is not valid UTF-8.
475
476  @return bool
477  */
```

```
function check_file_is_valid_UTF8(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_UTF8($s_string);
}//end check_file_is_valid_UTF8()



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
echo("sed\\\n");
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  --expression='s/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
```

```
  echo($s3."\n");
}
//*/
?>
```

3.68 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --add-test-files
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_fixed_point_build=""
```

```
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_fixed_point_build
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_add_test_files=0
  if [[ "$*" == *--add-test-files* ]]; then
    LFBFL_add_test_files=1
  fi
  readonly LFBFL_add_test_files
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    mkdir --parents -- --a/
    {
      printf "<?php\nfunction my_function(:;){\n\n}\n"
      # shellcheck disable=SC2016
      printf '$my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
      printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
    } > --a.php
    cp -- --a.php --a/
    {
      printf "from typing import Iterable\n"
      printf "from _collections_abc import dict_items\n"
      printf "\n"
      printf "from python_none_objects import NoneIterable\n"
      printf "\n"
      printf "def my_function(my_argument : int) -> None:\n"
      printf "  print(my_argument)\n"
      printf "\n"
      printf "my_function(\"abcd\")\n"
      printf "\n"
      printf "def my_function2(my_argument : int) -> None:\n"
      printf "    \"\"\"Some comment.\"\"\"\n"
      printf "    print(my_argument)\n"
      printf "\n"
      printf "def my_function3(my_argument : int) -> None:\n"
      printf "    \"\"\"Some comment.\"\"\"\n"
```

```
86        printf "\n"
87        printf "    print(my_argument)\n"
88        printf "\n"
89        printf "def my_function4(my_argument : int):\n"
90        printf "    \"\"\"Some comment.\"\"\"\n"
91        printf "    def my_sub_function():\n"
92        printf "        print(my_argument)\n"
93        printf "    return my_sub_function\n"
94        printf "\n"
95        printf "def my_function5(my_argument : int):\n"
96        printf "    \"\"\"Some comment.\"\"\"\n"
97        printf "\n"
98        printf "    def my_sub_function():\n"
99        printf "        print(my_argument)\n"
100       printf "    return my_sub_function\n"
101       printf "\n"
102     } > --a.py
103     cat -- --a.py
104     cp -- --a.py --a/
105     {
106       printf "def my_function(:;):\n\n\n"
107       printf 'my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
108       printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
109     } > --a2.py
110     cat -- --a2.py
111     cp -- --a2.py --a/
112     {
113       # Error 0 is missing shell header.
114       printf "my_function(){\n"
115       printf "   # Error 1: Indentation is not correct.\n"
116       printf "  sed -e 's/a/b/g' \\\\\n"
117       printf "      -e 's/a/b/g' \"\$1\" # Indentation is correct.\n"
118       printf "  my_variable=\"%%s\\n\"\n"
119       # shellcheck disable=SC2016
120       printf '  printf "${my_variable}"  # Error 2: shellcheck warning\n'
121       local LFBFL_ttp="ttp"
122       local LFBFL_URL="h${LFBFL_ttp}://example.com"
123       printf '  printf "%s" # Error 3: http -> https\n' "${LFBFL_URL}"
124       # shellcheck disable=SC2016
125       printf '  printf "${#some_'
126       printf 'arr_1}" # Error 4: no_size_of_array_first_element\n'
127       # shellcheck disable=SC2016
128       printf '  printf "${#'
129       printf 'arr_1}"  # Error 5: no_size_of_array_first_element\n'
130       printf '  if !'
131       # shellcheck disable=SC2016
```

```
      printf ' [[ $1 -eq 1 ]]; # Error 6: no_negation_before_bash_test\n'
      printf '  then  # Error 7: misplaced then\n'
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf "\n"
      # shellcheck disable=SC2016
      printf '  if [[ $1 -eq 1 ]];\n'
      printf '  then  # Error 8: another case of misplaced then\n'
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf '  my_variable="aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
      printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaa"  # Error 9 line too long.\n'
      printf '}\n'
    } > --a.sh
    cp -- --a.sh --a/
  fi
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    cat -- --a.py
    cat -- --a2.py
    rm --force --recursive -- --a/
    rm --force -- --a.php --a.py --a2.py --a.sh
  fi
}

upgrade_build_and_checks "$@"
```

3.69 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
      -- "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum -- "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    LFBFL_i_error=1
```

```
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"}\
      --output-document="$1.wget_sha512.temp"\
      -- "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum -- "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
    )
    if [[ "${LFBFL_check_download}" != "keep" ]]; then
      rm -- "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !